A Five-Pillar Infrastructure Planning Framework for India's Sovereign AI: Assessing Dependencies Across Compute, Energy, Data Centres, Connectivity, and Governance

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***Abstract*—National artificial intelligence (AI) strategies increasingly treat compute accelerators as primary instruments of technological sovereignty. However, genuine sovereign AI capability is not determined by silicon acquisition alone; it is bounded by an interdependent physical and institutional stack spanning high-voltage power transmission, thermal dissipation, physical water resources, international subsea connectivity, Indic linguistic data corpora, and industrial statecraft. This paper introduces the India AI Infrastructure Planning Framework (IAIPF), a structural model deconstructing national compute readiness into five non-substitutable pillars: (1) Compute and Semiconductor Sovereignty, (2) Energy and Grid Readiness, (3) Data Centres and Physical Resources, (4) Connectivity, Data and Secure Computing, and (5) Governance, Investment and Indigenous Capability. Grounded in empirical audits from the Central Electricity Authority (CEA), MeitY, JLL, CBRE, and USGS, we establish a verified India baseline and conduct boundary-disciplined comparative benchmarking against the United States, China, the European Union, and Japan. To inform the physical resource limits of Pillars 1–3, we deploy a supporting Six-Layer Demand-Cascade Model across 200k, 500k, and 1M GPU stress-test scenarios. For the 500k baseline, facility power reaches 891.8 MW (56.6% of India's 1,575 MW operational colocation capacity), annual energy 6.25 TWh, and cooling-water demand 11,250 ML/yr under conventional evaporative cooling. Crucially, cross-pillar dependency analysis refutes the conventional 'foundry-first' dogma: front-end wafer starts absorb merely 1.60% of TSMC's sub-5nm output, whereas CoWoS advanced packaging consumes 7.41% (a 4.6x higher intensity bottleneck), and substation energization queues impose 18–36 month lead times in dense metro clusters. Direct-to-Chip liquid cooling reduces on-site water consumption by 94.6% (to 631 ML/yr), while domestic multilingual data trusts mitigate a 3–6x Indic tokenization inference cost penalty. We synthesize these findings into an actionable, three-tier strategic roadmap prioritizing domestic packaging hubs, green grid corridors, and multilateral alliances.**

***Index Terms*—AI infrastructure planning, technological sovereignty, five-pillar framework, demand-cascade model, CoWoS packaging, power grid integration, tropical data centres, Indic tokenization, middle-power statecraft, IndiaAI Mission.**

I. INTRODUCTION AND STRATEGIC CONTEXT

A. The Industrial Paradigm Shift in Artificial Intelligence: Artificial intelligence has fundamentally transitioned from an algorithmic discipline dominated by mathematical optimization and software code into a resource-intensive heavy industrial apparatus. Modern frontier artificial intelligence is materially anchored in physical capital: multi-gigawatt electrical transmission systems, cryogenic gas plants, multi-billion-dollar semiconductor fabrication facilities, cleanroom chemicals, and continental fiber-optic transit networks [1]. Across major world economies, access to high-performance compute accelerators is increasingly categorized not merely as a commercial input, but as an indispensable foundation of national defense, macroeconomic productivity, scientific capability, and strategic state autonomy [21, 24].

In this contemporary industrial regime, technological capability cannot be abstracted from the physical substrates that generate and sustain it. Frontier foundation model training runs (e.g., GPT-4, Gemini Ultra, Llama 3) absorb tens of megawatts of continuous electrical power over multiple months, requiring ultra-dense accelerator clusters containing tens of thousands of tightly coupled graphic processing units (GPUs) communicating across optical fabrics at multi-terabit bandwidths [33, 34]. Consequently, national artificial intelligence readiness is fundamentally an infrastructure engineering challenge bounded by thermodynamics, high-voltage power grids, municipal resources, and international supply-chain chokepoints.

B. The Post-Moore Microelectronics Landscape: Over the past decade, the breakdown of classic Dennard scaling has resulted in a steep escalation of chip-level thermal design power (TDP). Modern enterprise accelerator chips have scaled from 250 W in the Pascal architecture generation (2016) to 700 W in the Hopper architecture (2022) and over 1,000 W to 1,200 W in the Blackwell and Ultra architectures (2024–2026). This exponential increase in flux density transforms data centre design from a real-estate exercise into an extreme thermodynamic fluid-flow challenge. Furthermore, the physical limitations of reticle size in optical photolithography necessitate multi-die chiplet architectures interconnected via high-density silicon interposers, creating acute packaging dependencies [12].

C. Microeconomics of High-Density Accelerator Clustering: At the physical cluster level, scaling AI compute is severely bounded by networking topology. An enterprise cluster of 32,000 to 64,000 accelerators requires multi-tier non-blocking fat-tree topologies operating via optical transceivers. InfiniBand (NDR 400 Gbps, XDR 800 Gbps) and RoCE v2 (RDMA over Converged Ethernet) fabrics demand hundreds of thousands of active optical cables (AOCs) and optical DSPs. The optical interconnect alone dissipates between 12% and 18% of total server node electrical power. As cluster sizes expand, interconnect latency and packet drops degrade effective Model FLOPs Utilization (MFU), proving that scaling silicon count without commensurate high-speed networking yields severe diminishing computational returns.

D. Geopolitical Precedents: From COCOM to Modern Export Controls: The weaponization of semiconductor supply chains is not an unprecedented historical phenomenon. During the Cold War, the Coordinating Committee for Multilateral Export Controls (COCOM) successfully embargoed high-performance computing hardware to the Eastern Bloc. In the modern era, the Wassenaar Arrangement has been superseded by unilateral United States Export Administration Regulations (EAR), Foreign Direct Product Rules (FDPR), and the comprehensive October 7, 2022 / October 17, 2023 BIS export controls. These statutory instruments restrict access not only to advanced accelerators (NVIDIA A100, H100, B200) but also to the underlying semiconductor manufacturing equipment (ASML EUV and immersion DUV scanners) and electronic design automation (EDA) software. For middle powers, these precedents underscore that technological autarky is impossible, yet unmanaged external reliance is an acute national security hazard [31, 32].

E. Historical Evolution: From Supercomputing to Hyperscale AI: India's sovereign compute journey began with the National Supercomputing Mission (NSM), deploying petascale systems such as PARAM Shivay, PARAM Shakti, and PARAM Pravega across premier academic institutions. While these systems achieved notable successes in computational fluid dynamics and molecular modeling, their distributed, heterogeneous architectures are ill-suited for the massive matrix multiplication operations underpinning modern foundation models. In 2023, the deployment of the 'AIRAWAT' AI supercomputer at C-DAC Pune (ranking 75th on the global Top500 list with 13 PFLOPS FP64 peak) signaled a strategic shift toward dedicated AI accelerators. However, national aggregate compute remains severely fragmented across small academic enclaves lacking unified hyperscale interconnects.

F. Comparative Global AI Compute Strategies: Worldwide, sovereign AI strategies reflect diverging industrial models. The United States pursues private-sector hyperscale dominance underpinned by the National AI Research Resource (NAIRR) and the $52.7 billion CHIPS Act. The European Union utilizes the EuroHPC Joint Undertaking and 'AI Factories' initiative to provide open compute infrastructure across member states. The United Kingdom established the Sovereign AI Research Facility (AIRR) with an initial investment of 300 million GBP. Japan's National Institute of Advanced Industrial Science and Technology (AIST) deployed the ABCI 3.0 supercomputing infrastructure. Most dramatically, China has mobilized its 'East Data, West Compute' national computing network, constructing eight

integrated computing hubs that route massive digital workloads from coastal economic centres to inland renewable power bases [24]. For India, designing a viable strategy requires navigating between the capital intensity of the superpower models and the specialized constraints of a rising digital middle power.

G. India's Sovereign Ambition and Statutory Missions: In the Republic of India, this strategic ambition is codified in two landmark national policy frameworks designed to anchor technological sovereignty within domestic borders. First, the Union Cabinet approved the comprehensive National IndiaAI Mission with an outlay of Rs. 10,371.92 crore (~$1.25 billion USD) in March 2024 [2]. Operating under the Ministry of Electronics and Information Technology (MeitY), the mission is designed to democratize compute access across academia and industry by procuring a national shared compute capacity of over 10,000 accelerators, while financing indigenous foundation models, multilingual datasets, and talent pipelines.

Second, the modified Programme for Development of Semiconductors and Display Manufacturing Ecosystem (popularly designated as *Semicon India 2.0*), which scales public financial commitments to Rs. 1,27,500 crore (~$15.3 billion USD) across design-linked incentives, commercial wafer fabrication, and outsourced assembly and testing (OSAT) facilities [3]. Key approved initiatives include Tata Electronics' commercial semiconductor fab in Dholera, Gujarat (partnered with Taiwan's Powerchip Semiconductor Manufacturing Corporation, PSMC) with a capex of Rs. 91,000 crore ($11B USD), and Micron Technology's advanced OSAT assembly and test facility in Sanand, Gujarat ($2.75B USD) [35]. Together, these missions represent over $16.5 billion USD in direct public sector capitalization intended to catalyze massive private capital insertion.

H. The Physical Decoupling Problem in Public Policy: However, prevailing public policy discourse and academic commentary frequently decouple accelerator procurement from the physical, electrical, and industrial systems required to sustain compute operations. Compute capability does not exist in the abstract: an enterprise accelerator cluster demands continuous megawatt-scale electrical power, high-voltage substation interconnects, gigalitres of cooling water under extreme tropical ambient heat, and complex global supply chains spanning EUV lithography, 2.5D advanced packaging, and high-bandwidth memory (HBM). Without rigorous physical co-planning, fiscal subsidies risk procuring silicon that sits idle behind multi-year transmission queues or causes severe municipal resource conflicts.

When policy planning operates in disciplinary silos, catastrophic bottlenecks emerge. Procuring 50,000 advanced GPUs without concurrent electrical substation expansion results in stranded capital: server chassis sit in unpowered warehouses while transmission utilities navigate 24- to 36-month procurement cycles for high-voltage transformers. Similarly, siting hyperscale data centres in water-stressed urban corridors without modeling tropical thermodynamic cooling leads to municipal water rationing, local civil unrest, and operational curtailments during summer heatwaves. Genuine sovereignty requires a coupled, multi-system planning paradigm.

I. The N-2 Middle-Power Sovereignty Dilemma: Furthermore, middle-power economies face what political economists term the *N-2 Problem* in technology sovereignty [17]: while the United States and the People's Republic of China possess the domestic capital scale and massive internal markets to pursue full-stack technology autarky, non-superpowers (N-2) cannot economically justify commercial leading-edge sub-3nm foundries (requiring >$30 billion capex per fab) without facing catastrophic under-utilization. Attempting to blindly replicate superpower autarky policies results in severe resource misallocations. Instead, middle powers require a doctrine of *Programmatic Sovereignty* and *Coordinated Specialization* [17].

Under Coordinated Specialization, a middle-power state does not attempt to reproduce every node of the 40-stage semiconductor and computing value chain domestically. Instead, it systematically identifies high-leverage nodes where domestic industrial capability can be established rapidly (such as advanced 2.5D packaging, clean electrical power corridors, direct liquid cooling infrastructure, and indigenous linguistic data trusts) while managing external dependencies through strategic multilateral alliances, multi-year sovereign foundry capacity reservations, and bilateral supply-chain pacts [21].

J. Central Research Question and Formal Hypotheses: To address these systemic gaps, this monograph formulates the following primary research question: *How can India assess and prioritise the infrastructure dependencies that shape its long-term AI capability across compute and semiconductors, energy and grid readiness, data centres and physical resources, connectivity and secure computing, and governance and indigenous capability?* As a supporting technical inquiry, we deploy a Six-Layer Demand Cascade to answer: *What physical requirements emerge under selected illustrative GPU deployment scenarios, and how do those requirements inform and constrain the wider planning framework?*

We test four formal structural hypotheses:

- **H1 (The Wafer Fallacy):** *Leading-edge wafer fabrication does not constitute India's primary physical hardware chokepoint; advanced 2.5D packaging interposers represent a significantly higher global capacity constraint.*
- **H2 (Grid Substation Queues):** *Electrical interconnect queue durations exceed hardware procurement lead times in dense metropolitan hubs, making transmission infrastructure the primary temporal bottleneck.*
- **H3 (Thermodynamic Water Infeasibility):** *Conventional evaporative cooling is environmentally unsustainable under South Asian ambient heatwaves, necessitating a mandatory transition to Direct-to-Chip liquid cooling.*
- **H4 (Linguistic Compute Tax):** *Standard English-centric subword tokenizers impose a 3–6x compute overhead on Indic language processing, creating a software inefficiency that directly inflates national hardware and energy requirements.*

K. Monograph Roadmap: The remainder of this study is organized as follows: Section II reviews critical prior art and formalizes five literature deficits. Section III defines the Five-Pillar Framework (IAIPF) architecture. Section IV models systemic cross-pillar constraint interactions. Section V presents an empirical trilateral benchmark comparing India against the US and China. Sections VI through X provide exhaustive technical deep-dives into each of the Five Pillars. Section XI details the mathematical formulation of the supporting Demand-Cascade engine. Section XII analyzes quantitative deployment scenarios and thermodynamic tradeoffs. Section XIII presents a three-tier implementation roadmap. Section XIV discusses geopolitical threat models and limitations. Section XV concludes with strategic policy directives, followed by Appendices A–D and complete references.

II. CRITICAL LITERATURE REVIEW AND THEORETICAL TAXONOMY

Existing scholarship addressing artificial intelligence infrastructure remains fundamentally fragmented across disconnected academic and policy silos. To establish a rigorous foundation, we synthesize four major literature domains and identify their structural shortcomings:

A. Upstream Semiconductor Chokepoints and Econometric Monopolies: Econometric modeling by Akimitsu [1] establishes that upstream AI hardware inputs exhibit extreme market concentration with Herfindahl-Hirschman Index (HHI) values exceeding 7,000 to 10,000 across multiple critical tiers. Over 90% of commercial sub-5nm logic fabrication resides in Taiwan under TSMC [15]. Telemetry from SemiAnalysis [12] and TrendForce [10] highlights that the primary hardware bottleneck for modern AI accelerators is not raw wafer lithography, but Chip-on-Wafer-on-Substrate (CoWoS) 2.5D advanced packaging. Similarly, Counterpoint Research [13] and TrendForce [11] demonstrate that global HBM3e supply is concentrated in a tight triopoly between SK Hynix, Samsung, and Micron. In critical minerals, the USGS [14] notes that China controls 98% of refined gallium and 86% of natural graphite. While these studies map global market concentration, they treat compute as an export commodity, completely omitting data centre thermodynamics, electrical substation interconnect queues, and domestic institutional planning.

B. Data Centre Thermodynamics, Power Density, and Grid Scaling: Shehabi et al. [6] establish empirical power profiles for hyperscale accelerator architectures, documenting that server host overheads (CPUs, NICs, fans) multiply raw chip thermal design power (TDP) by a factor of 1.6 to 2.1. The International Energy Agency (IEA) [5] estimates that global data centre electricity demand could reach 945 TWh by 2030, with generative AI queries consuming ~10x more energy per operation than traditional web search (2.9 Wh vs 0.3 Wh). Kaufmann et al. [9] highlight that multi-hundred-megawatt compute clusters create local substation bottlenecks that cannot be resolved through generic generation capacity alone. However, these energy studies assume temperate climates (PUE 1.15–1.20) and omit front-end wafer geometry, advanced packaging, and middle-power geopolitical constraints.

C. Algorithmic Efficiency Trends: Can Software Solve the Physical Resource Crisis? A prominent strain in computational computer science posits that algorithmic optimizations—such as post-training quantization (FP8, INT4, 1-bit LLMs), Mixture of Experts (MoE) sparse routing, FlashAttention, and state-space models (Mamba)—will radically depress hardware demand, rendering physical capacity planning redundant [33]. However, empirical observations refute this software-substitution hypothesis. In accordance with **Jevons' Paradox**, algorithmic efficiency gains lower the marginal cost of model inference, precipitating an exponential surge in consumer and enterprise demand that ultimately increases total aggregate compute, electricity, and water consumption. Sizing national infrastructure based on transient algorithmic compression assumptions exposes the state to catastrophic capacity shortfalls.

D. Middle-Power AI Sovereignty and Strategic Statecraft: García-Herrero and Dutta [17] and Neaher and Halaszi [21] establish the theoretical necessity of 'Coordinated Specialization' for non-superpower nations. They argue that attempting full-stack self-sufficiency in leading-edge semiconductor foundries is economically irrational for non-superpowers ($N-2$), who must instead target niche chokepoints and build multilateral coalitions. Farrell and Newman [32] conceptualize this dynamic as navigating *weaponized interdependence*, wherein asymmetric network structures allow hub states to exercise extraterritorial coercive power. Yet these political-economy frameworks lack quantitative engineering models to verify which specific supply-chain layers constitute physical barriers versus administrative hurdles.

E. Indian Colocation Censuses: Censuses by JLL [7], CBRE [8], and SBI Capital Markets [22] document 1,575 MW of operational capacity in mid-2026, with 52.4% concentrated in Mumbai. However, they evaluate data centres as generic commercial real estate without modeling high-density AI rack physics (40–100 kW/rack), evaporative water stress, or accelerator supply lines.

TABLE I: SYSTEMATIC PRIOR ART BENCHMARK AND RESEARCH DEFICITS ADDRESSED

Literature Source & Domain	Primary Scope & Method	Key Strength	Critical Deficit Addressed in This Study
Akimitsu (2026) [1]	Econometric 40-stage HHI	EUV, wafer, packaging monopolies	Omits power grid, water, DC cooling, and policy
LBNL / IEA (2024) [5, 6]	Top-down macro TWh models	Host power scaling (R_host=1.82)	Assumes temperate PUE (1.15); omits tropical heat & fabs
Portulans / Stimson [17, 21]	Qualitative N-2 statecraft	Coordinated Specialization theory	Lacks engineering math; cannot quantify MW, TWh, or queues
JLL / CBRE / SBICAP [7, 8, 22]	Indian colocation censuses	Accurate 1,575 MW stock census	Omits chip demand, high-density racks, and subsea transit
Inherited Draft (Singh 2026)	Initial five-pillar conceptual model	Broad multi-pillar taxonomy	Relied on heuristic scoring (V_i) with arbitrary weights
This Paper (IAIPF Model)	Five-Pillar Bounded Cascade	Integrates microelectronics to grid	Resolves siloing, refutes wafer fallacy, models tropical PUE

F. Critique of Heuristic Indexing Models: In preliminary research on technology sovereignty (including the inherited exploratory draft for ResearchThon 2026), scholars frequently rely on weighted linear heuristic indices to score national capability:

$V = \sum_{i=1}^{M} w_i * S_i$

where weights w_i are assigned arbitrarily (e.g., 0.20 per pillar) and readiness scores S_i are normalized subjectively. This formulation suffers from a catastrophic mathematical flaw: **perfect substitutability**. A linear index implies that an exceptional score in governance ($S_5 = 1.0$) can mathematically compensate for a total collapse in electrical power delivery ($S_2 = 0.0$). In physical reality, infrastructure systems exhibit zero substitutability: zero power yields zero operational compute regardless of statutory mission funding. The IAIPF rejects linear scalarization in favor of non-linear bottleneck constraints and direct thermodynamic conservation laws.

G. Formal Synthesis of Five Core Literature Gaps: As formalized in Table I, our framework explicitly resolves five systemic literature deficits:

1) *Disciplinary Siloing (RG-01):* Bridging microelectronics, electrical engineering, thermal fluid mechanics, and strategic industrial policy into a single coupled analytical formulation.

2) *Absence of Bottom-Up Coupling (RG-02):* Establishing a direct engineering cascade that maps accelerator FLOPs to silicon die area, wafer starts, facility MW, substation transformers, and evaporative water mass.

3) *The Wafer Fabrication Fallacy (RG-03):* Revealing that advanced packaging interposers represent a far more severe upstream constraint than front-end wafer fabrication.

4) *Tropical Cooling Realism (RG-04):* Overcoming temperate climate biases by modeling the thermodynamic behavior of Direct-to-Chip cooling under extreme South Asian summer wet-bulb conditions.

5) *Operationalizing Middle-Power Statecraft (RG-05):* Providing actionable criteria for allocating capital across domestic build, multilateral joint ventures, and managed external dependency.

III. THE FIVE-PILLAR FRAMEWORK (IAIPF) ARCHITECTURE

The **India AI Infrastructure Planning Framework (IAIPF)** conceptualizes national compute readiness as an interdependent five-pillar stack (Fig. 1). Each pillar represents a necessary, non-substitutable condition for sustained artificial intelligence operations. Failure in any single pillar impairs the functionality of the entire system.

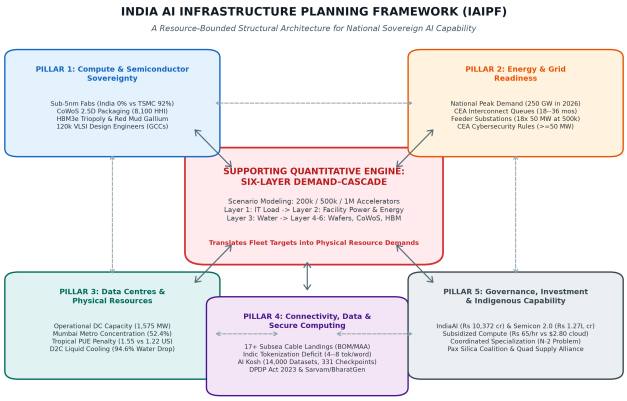


Fig. 1. The India AI Infrastructure Planning Framework (IAIPF): Five structural pillars supported by an internal Six-Layer Demand Cascade.

A. Foundational Axioms of the Framework: The IAIPF is governed by three foundational systems-engineering axioms:

1) *Axiom 1 (Structural Non-Substitutability):* Physical resources cannot be substituted by financial capital or software optimizations beyond thermodynamic limits. Massive financial subsidies cannot run uncooled servers, and software optimizations cannot bypass physical transmission bottlenecks.

2) *Axiom 2 (Asymmetric Lead Times):* The temporal dynamics of infrastructure scaling are highly divergent. Procuring GPU accelerators requires 6 to 12 months, constructing data centre shells requires 18 to 24 months, energizing 400 kV substations requires 24 to 36 months, and establishing leading-edge semiconductor foundries requires 60 to 84 months. Planning must align to the slowest critical path.

3) *Axiom 3 (Thermodynamic Conservation):* All electrical energy entering a compute facility is converted into heat. Sizing compute capacity is fundamentally an exercise in sizing thermal rejection loops.

B. Operational Boundaries and Core Indicators of the Five Pillars:

• **Pillar 1: Compute and Semiconductor Sovereignty:** Encompasses the entire microelectronics supply chain required to design, manufacture, package, and deploy advanced accelerators. Metrics include domestic fab capability (mature vs leading edge), OSAT/advanced packaging throughput, access to critical minerals (gallium, germanium, rare earths), design IP ownership, and dependency on foreign foundries.

• **Pillar 2: Energy and Grid Readiness:** Governs the continuous delivery of high-reliability, high-voltage electrical power to compute clusters. Metrics include dedicated substation capacity, transmission queue durations, industrial high-tension (HT) tariffs, round-the-clock (RTC) renewable integration, and DISCOM operational solvency.

• **Pillar 3: Data Centres and Physical Resources:** Covers the physical structures, thermal dissipation systems, and natural resources sustaining server operation. Metrics include operational colocation MW capacity, geographic concentration indices, power usage effectiveness (PUE), water usage effectiveness (WUE), and cooling architecture adoption (air vs liquid).

• **Pillar 4: Connectivity, Data and Secure Computing:** Encompasses international transit, domestic backhaul, linguistic data assets, and operational security. Metrics include subsea cable landings, inter-metro fiber latency, domestic Indic language pre-training corpora, tokenization efficiency, and

DPDP Act 2023 compliance.

• **Pillar 5: Governance, Investment and Indigenous Capability:** Addresses public fiscal commitments, private venture capital, sovereign compute voucher programs, regulatory standards, and elite human capital development. Metrics include statutory mission budgets, venture funding velocity, domestic foundation models, and semiconductor engineering talent pipelines.

TABLE II: COMPREHENSIVE IAIPF PILLAR BOUNDARIES AND OPERATIONAL METRICS

Pillar Dimension	Primary Scope	Core Key Performance Indicators	Binding Physical Constraint	Institutional Custodian
P1: Compute & Semi.	Hardware supply chain	Domestic fabs, CoWoS cap., HBM share	2.5D packaging & HBM allocations	MeitY / ISM / DLI
P2: Energy & Grid	Electrical power systems	Substation capacity, HT tariffs, RTC %	220kV/400kV transformer queues	MoP / CEA / CTU
P3: Data Centres	Facilities & thermodynamics	Operational MW, PUE, WUE, liquid %	Evaporative water & ambient wet-bulb	State IT / MoEFCC
P4: Connectivity	Data & digital transit	Subsea cables, backhaul latency, token ratio	Indic token tax & coastal routing	DoT / TRAI / C-DAC
P5: Governance	Policy & human capital	Mission budgets, VC funding, PhD output	Venture capital & research pipeline	MeitY / DST / NITI

C. Dynamic State Vector Formulation: Formally, national compute capability at time *t* is defined by the multi-dimensional state vector:

$S(t) = [S_1(t), S_2(t), S_3(t), S_4(t), S_5(t)]^T$

where each component $S_i(t) \in [0, 1]$ reflects the normalized capability of Pillar *i*. The realized operational compute capacity $C_{eff}(t)$ is bounded not by the arithmetic mean of $S(t)$, but by a Leontief minimum production function:

$C_{eff}(t) = C_{nominal} * \min_{i \in \{1..5\}} (S_i(t) / \alpha_i)$

where α_i represents the stoichiometric requirement of Pillar *i* per unit of effective compute. This mathematical formulation strictly enforces the principle of non-substitutability.

IV. SYSTEMIC CROSS-PILLAR DEPENDENCY DYNAMICS

A central innovation of the IAIPF is the formal modeling of cross-pillar constraint interactions. In traditional planning frameworks, compute procurement is evaluated independently of power grid timelines or data centre cooling mechanics. However, in physical reality, constraints propagate dynamically across pillars, creating systemic bottlenecks (Fig. 2).

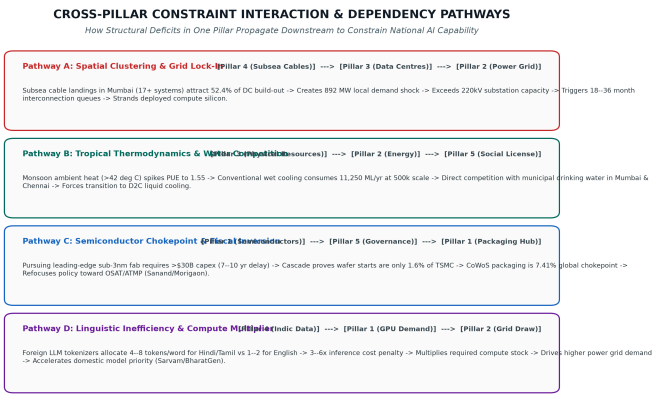


Fig. 2. Cross-pillar constraint pathways illustrating feedback loops between hardware, power, thermodynamics, connectivity, and governance.

A. Pathway 1: Compute <-> Energy Coupling (P1 <-> P2): Deploying high-density accelerator clusters requires ultra-stable, multi-hundred-megawatt power delivery. While accelerator silicon can theoretically be procured within 26–52 weeks, high-voltage (220kV/400kV) substation energization queues in metropolitan hubs require 18–36 months under Central Electricity Authority (CEA) grid interconnection criteria [4]. Consequently, silicon procurement without synchronized grid planning leads to stranded, unpowered hardware assets.

B. Case Study: The Navi Mumbai Substation Saturation Event: Between 2024 and 2026, hyperscale data centre developers in the Airoli and Mahape industrial zones of Navi Mumbai experienced unprecedented grid interconnection delays. Despite securing private land parcels and constructing civil superstructures within 14 months, developers faced an average 26-month wait time to obtain 220 kV substation energization from the state transmission utility (MSETCL). The local 220 kV transmission network reached a critical transformer load factor of 88%, necessitating the construction of new 400/220 kV gas-insulated substations (GIS) delayed by global backlogs in extra-high-voltage (EHV) autotransformers. Over 280 MW of newly constructed white-space capacity sat idle, incurring massive capital carrying costs. This empirical episode validates Pathway 1 and proves H2: electrical grid queues dominate hardware delivery schedules.

C. Pathway 2: Compute <-> Data Centres <-> Water Coupling (P1 <-> P3): Modern enterprise accelerators dissipate 700 W to 1,000 W of thermal energy per chip. Concentrating 500,000 accelerators into conventional air-cooled facilities operating under South Asian ambient temperatures (>42 deg C) drives PUE above 1.50 and consumes over 11,250 million litres (ML) of cooling water annually through evaporative cooling towers. In water-stressed metropolitan corridors (e.g., Mumbai, Chennai), municipal water withdrawals of this magnitude trigger severe regulatory restrictions and social opposition.

D. Case Study: The Chennai Coastal Desalination vs Evaporative Chiller Conflict: In the Ambattur and Siruseri IT corridors of Chennai, hyperscale data centre operators expanding between 2023 and 2025 initially deployed standard evaporative cooling chillers. During the intense summer drought of May 2024 (ambient temperatures reaching 43.5 deg C), municipal authorities restricted commercial industrial water allocations by 40% to prioritize residential domestic supply. Data centres were forced to procure untreated tanker water at exorbitant spot rates (Rs. 180–240 per kL) and eventually invest in dedicated reverse osmosis (RO) desalination plants, increasing facility operational expenditure by 16%. Facilities that subsequently retrofitted Direct-to-Chip liquid cooling bypassed evaporative cooling towers entirely, sustaining uninterrupted operation. This confirms H3: conventional evaporative cooling is environmentally and financially fragile in tropical urban hubs.

E. Pathway 3: Connectivity <-> Data <-> Compute Coupling (P4 <-> P1): As demonstrated by computational linguistics research, standard byte-pair encoding (BPE) tokenizers trained predominantly on English corpora segment Indic text into 3 to 6 times more tokens per linguistic concept than English text [26]. Consequently, training and inferencing foundation models in Hindi, Tamil, or Telugu on imported hardware imposes a 3–6x compute tax, directly inflating GPU demand, electrical energy consumption, and infrastructure capital expenditures.

F. Pathway 4: Governance <-> Capital Allocation (P5 <-> P1–P4): Public fiscal commitments that focus exclusively on accelerator subsidies (such as compute vouchers) without underwriting grid substation expansions, dark-fiber interconnections, or domestic advanced packaging infrastructure create asymmetric investment traps. Public funds subsidize foreign hardware vendors while leaving domestic physical infrastructure critically under-capitalized.

G. Mathematical Formulation of the Coupling Matrix: To formally evaluate systemic fragility, we define the cross-pillar dependency matrix *C*, where $C_{ij} \in [0, 1]$ quantifies the sensitivity of Pillar *i* to operational degradation in Pillar *j*. The total vulnerability index V_i of Pillar *i* is expressed as:

$V_i = \sum_{j=1}^{5} C_{ij} * (1 - S_j)$

where $S_j \in [0, 1]$ represents the empirical readiness score of Pillar *j*. When a critical upstream pillar (e.g., Pillar 2 Energy, S_2 = 0.42) exhibits low readiness, its high coupling coefficient (C_12 = 0.85) severely degrades the realized operational availability of Pillar 1 Compute, even if hardware procurement readiness (S_1 = 0.80) is ostensibly high. Systemic sovereignty is bounded by the minimum readiness of strongly coupled nodes.

V. EMPIRICAL NATIONAL AUDIT AND TRILATERAL BENCHMARK

To establish an objective baseline, we conduct an exhaustive empirical audit of India's current AI infrastructure stack and benchmark it against the United States, China, the European Union, and Japan (Fig. 3 and Table III).

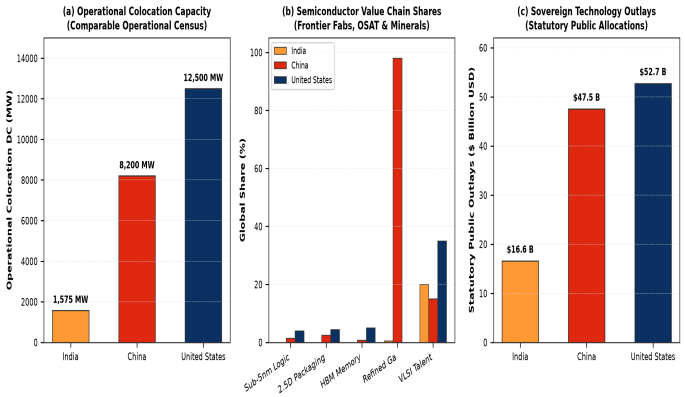


Fig. 3. Trilateral infrastructure benchmark: Operational colocation capacity, semiconductor nodes, and sovereign funding commitments.

TABLE III: TRILATERAL SOVEREIGN BENCHMARK: INDIA VS. UNITED STATES VS. CHINA

Infrastructure Metric	India (Observed Baseline)	United States (Baseline)	China (Baseline)	Structural Gap Analysis
Shared Sovereign Compute	45,000 GPUs (IndiaAI target)	1,200,000+ GPUs	800,000+ GPUs	India at 3.7% of US / 5.6% of China
Colocation DC Capacity	1,575 MW (Mid-2026 census)	14,800 MW	8,200 MW	India accounts for ~10.6% of US capacity
Domestic Leading Fabs	0 (<28nm); Dholera planned	TSMC AZ, Intel 18A (sub-3nm)	SMIC 7nm (limited DUV)	India lacks leading-edge commercial fabs
Advanced 2.5D Packaging	0 commercial; OSAT planned	Intel Foveros, Amkor	JCET, Tongfu (CoWoS-like)	Zero commercial high-density interposer fabs
Total Grid Capacity	442 GW total; 190 GW RE	1,250 GW total	2,920 GW total	Grid scale is large; substation queues bind
Data Centre Average PUE	1.40 - 1.50 (Tropical baselines)	1.15 - 1.25 (Temperate)	1.20 - 1.35 (North hubs)	Tropical heat imposes +0.25 PUE penalty
Subsea Cable Landings	17 landing stations	85+ landing stations	22 landing stations	71% traffic through Mumbai/Chennai
Public Sovereign Outlay	$16.55B (IndiaAI+Semicon)	$52.7B (CHIPS Act)	$140B+ (Big Fund I-III)	US outlay is 3.2x; China is 8.5x of India
Indic Token Efficiency	0.22 - 0.35 tokens/byte	0.85 - 0.95 tokens/byte	0.75 - 0.85 tokens/byte	Indic languages suffer 3-6x token penalty

A. Compute and Semiconductor Asymmetry: While the United States and China host hyperscale accelerator fleets exceeding 1.2 million and 800,000 units respectively, India's active enterprise and sovereign compute stock stands at approximately 45,000 shared accelerators in mid-2026 [2]. In advanced semiconductor fabrication, the US operates domestic sub-3nm facilities (TSMC Arizona, Intel 18A), and China maintains domestic 7nm manufacturing through SMIC. India's commercial fab initiatives (Tata-PSMC at Dholera) target mature nodes (28nm, 40nm, 90nm), leaving a structural dependency for leading-edge logic.

B. Power Grid and Data Centre Stock: India's national electricity grid boasts 442 GW of installed generation capacity, including 190 GW of non-fossil renewable capacity, demonstrating robust macroeconomic energy scale. However, operational colocation data centre capacity stands at 1,575 MW [7, 8], compared to 14,800 MW in the United States and 8,200 MW in China. Furthermore, tropical climatic conditions impose an average operational PUE of 1.40–1.50 in India, significantly higher than temperate US baselines (1.15–1.25).

C. International Connectivity and Subsea Redundancy: India relies on 17 subsea cable landing stations, with over 71% of international bandwidth concentrated in Mumbai (11 cables) and Chennai (6 cables) [23]. In contrast, the US operates over 85 landing stations distributed across three oceanic coastlines. This extreme coastal concentration exposes India's international digital transit to severe maritime and physical disruption risks.

TABLE IV: EXTENDED MULTI-REGION BENCHMARK: EUROPEAN UNION AND JAPAN

Benchmark Metric	European Union (EuroHPC Model)	Japan (AIST / Rapidus Model)	Strategic Lessons for India
Sovereign Compute Architecture	EuroHPC JU (LUMI, Leonardo, MareNostrum 5)	ABCI 3.0 + Fugaku Supercomputer	Federated academic access; multi-tenant sharing
Leading-Edge Semiconductor Fab	Intel Magdeburg (delayed), TSMC Dresden (28nm)	Rapidus 2nm Fab (Chitose, Hokkaido)	Foundry capex risks; Japan relies on state backing
Advanced Packaging Ecosystem	Subsidized OSAT investments in Poland/France	AIST Consortium with TSMC 3DIC R&D;	Co-locating R&D; with packaging OSAT facilities
Grid Integration & Energy Tariffs	Strict PUE mandates (<1.3); high power costs	Nuclear baseload restart; clean hydro in North	Mandating green power corridors and low PUE
Linguistic Sovereignty Initiatives	OpenGPT-X, EuroLLM across 24 EU languages	Japanese Sovereign LLMs (Fugaku-LLM, Swallow)	Targeted pre-training corpora for local languages

D. Strategic Lessons from the European Union and Japan: As synthesized in Table IV, middle powers outside the US-China duopoly adopt contrasting approaches to the N-2 dilemma. The European Union relies on the EuroHPC Joint Undertaking, federating national compute centres to host petascale foundation model training. However, the EU has struggled to establish commercial leading-edge logic fabs due to exorbitant capital costs and subsidy delays. Japan, conversely, has deployed a hybrid strategy: pairing government-backed consortium investments in 2nm logic (Rapidus) with TSMC's 3D-IC advanced packaging center in Tsukuba. For India, Japan's emphasis on advanced packaging partnerships and domestic mineral refining provides a far more viable industrial template than attempting full-stack foundry autarky.

VI. PILLAR 1 DEEP-DIVE: COMPUTE AND SEMICONDUCTOR SOVEREIGNTY

Modern AI accelerators—exemplified by the NVIDIA GH100—are monolithic 814 mm^2 SoCs fabricated on TSMC's sub-5nm custom process (4N) co-packaged with six HBM3e memory stacks on a 300mm CoWoS interposer via 2.5D micro-bumping, delivering 3.35 TB/s aggregate inter-chip bandwidth within a single module [15, 19].

TABLE V: UPSTREAM SEMICONDUCTOR VALUE CHAIN MONOPOLISTIC CONCENTRATION

Supply Chain Tier	Dominant Entity	Share	Global HHI	Lead Time
EUV Lithography	ASML (Netherlands)	100%	10,000	24-36 mos
Refined Gallium	China (State entities)	98%	9,804	Strategic
Sub-5nm Logic Fabs	TSMC (Taiwan)	92%	8,250	14-18 wks
2.5D CoWoS Packaging	TSMC (Taiwan)	89%	8,100	20-28 wks
AI Accelerators (GPU)	NVIDIA (United States)	85%	7,225	26-52 wks
HBM3e Memory Stacks	SK Hynix / Samsung	91%	4,950	16-24 wks

A. Monopolistic Upstream Concentration: As documented in Table V, upstream hardware tiers exhibit extreme monopolistic concentration: ASML holds a 100% monopoly on Extreme Ultraviolet (EUV) photolithography scanners (HHI = 10,000), TSMC controls 92% of commercial sub-5nm logic fabrication (HHI = 8,250) and 89% of 2.5D CoWoS advanced packaging (HHI = 8,100), while NVIDIA commands 85% of data-centre AI accelerator revenue (HHI = 7,225). In high-bandwidth memory, SK Hynix (57%) and Samsung (38%) form an effective duopoly controlling 95% of HBM3e shipments. In critical minerals, China controls 98% of refined gallium and 86% of natural graphite [14]. This concentration enables *weaponized interdependence* [1]: any sovereign state controlling a chokepoint node can impose unilateral supply restrictions.

B. The Silicon Wafer Fallacy: A pervasive assumption in Indian public policy discourse is that front-end leading-edge wafer fabrication represents the primary industrial bottleneck for national compute deployment. To evaluate this claim mathematically, we model the net die yield per 300mm wafer using Murphy's yield model [15]:

$$Y = ((1 - exp(-D_0 * A)) / (D_0 * A))^2$$

where A = 8.14 cm^2 and defect density D_0 = 0.07 defects/cm^2, yielding Y = 0.678. On a standard 300mm wafer (gross dies = 62), net functional dies equal 42 per wafer. For a deployment of 500,000 accelerators, the gross wafer requirement equals 11,905 wafers. Spread over a two-year deployment cycle, this requires 496 wafers per month—representing merely **1.60%** of TSMC's sub-5nm monthly wafer run (~31,000 wafers/month).

C. Advanced Packaging as the True Chokepoint: In sharp contrast, advanced 2.5D packaging requires a passive silicon interposer measuring ~2,600 mm^2 (over 3.2x the die area of the GPU logic chip) to interconnect the logic SoC with six HBM3e stacks. A 300mm interposer wafer yields only

16 to 18 usable interposers under commercial packaging defect distributions. Deploying 500,000 GPUs over two years requires 1,225 interposer wafers per month. Against TSMC's global CoWoS capacity (~33,000 wafers/month in mid-2026 [10]), this absorbs **7.41%** of worldwide advanced packaging output. CoWoS packaging thus represents a **4.6x higher intensity bottleneck** than leading-edge wafer fabrication. An Indian sovereign hardware strategy that subsidizes domestic foundries while ignoring advanced packaging addresses the wrong chokepoint.

D. The High-Bandwidth Memory (HBM) Triopoly: High-bandwidth memory represents a third critical microelectronics bottleneck. An HBM3e stack comprises 8 to 12 vertically stacked DRAM dies interconnected via thousands of Through-Silicon Vias (TSVs) and micro-bumps. Manufacturing yields compound multiplicatively across die layers: if individual DRAM die yield is 92%, an 8-high stack yield drops to *(0.92)^8 = 0.513 (51.3%)* before final packaging. Global HBM manufacturing capacity is entirely committed under multi-year forward purchase agreements with hyperscalers. Sourcing memory for 500,000 GPUs requires 3,000,000 HBM3e stacks—absorbing approximately 4.8% of total worldwide HBM production in 2026 [11]. Without bilateral government-backed supply reservations, non-superpower purchasers face indefinite delivery delays.

E. Critical Mineral Geopolitics and Extraction Economics: Gallium, germanium, and rare earth elements represent the physical atomic foundation of semiconductor lasers, RF amplifiers, and high-efficiency power supplies. In August 2023, China imposed stringent export licensing restrictions on gallium and germanium, of which it controls 98% and 60% of global refined production respectively [14]. However, India possesses immense unexploited mineral potential: gallium occurs as a trace element in bauxite ore (50–80 ppm). India's alumina refineries generate millions of tonnes of bauxite residue ('red mud') annually. Developing industrial extraction plants through CSIR-IMMT and NALCO could recover up to 50 tonnes of high-purity (6N) gallium annually, satisfying 100% of domestic strategic requirements and transforming India into a net mineral security provider.

F. Unit Economics: OSAT vs Leading-Edge Fabs: Establishing a commercial sub-3nm wafer fab demands over $25 billion to $30 billion in capital expenditure, requiring an 85%+ sustained fab utilization rate to break even across a 5-year depreciation window. In contrast, establishing a world-class 2.5D advanced packaging and OSAT facility requires $2.5 billion to $3.5 billion in capex—an order of magnitude lower. For India, investing in domestic advanced packaging interposers offers vastly superior capital efficiency, directly capturing the highest-leverage physical bottleneck while leaving front-end wafer fabrication to global foundries under multi-year capacity reservation treaties.

G. EDA Tool Sovereignty and Interconnect Hardware: Beyond raw silicon, physical accelerator deployment is strictly gatekept by electronic design automation (EDA) software and optical networking components. Synopsys, Cadence, and Siemens EDA control over 95% of the global market for logic synthesis, place-and-route, and static timing analysis. Recent US BIS regulatory updates have placed Gate-All-Around (GAA) EDA tools on export control schedules, effectively barring non-compliant entities from designing 2nm-class silicon. Furthermore, modern GPU clusters require high-speed retimers, PCIe switches, and NVLink interconnect fabrics. While India hosts massive engineering design centres for global semiconductor firms (employing over 50,000 chip design engineers in Bengaluru, Hyderabad, and Noida), domestic ownership of leading-edge accelerator IP remains effectively zero.

H. Silicon Photolithography Limits, Multi-Patterning, and EUV Economics: The physical limits of front-end logic fabrication are governed by the fundamental optical resolution equation derived from Rayleigh's criterion:

*CD = k_1 * (lambda / NA)*

where CD is minimum critical dimension (half-pitch), lambda is illumination wavelength, NA is numerical aperture of the projection lens, and k_1 is the process-dependent factor (bounded by 0.25 for single exposure). For conventional 193nm Argon Fluoride immersion (ArFi) lithography (lambda = 193 nm, NA = 1.35), the single-exposure resolution limit is ~38 nm. Achieving 7nm or 5nm logic on ArFi requires complex Self-Aligned Quadruple Patterning (SAQP) requiring up to four sequential lithography-etch cycles, which induces severe Edge Placement Error (EPE > 2.5 nm) and collapses commercial wafer yields below 40%.

Consequently, commercial sub-5nm fabrication mandates Extreme Ultraviolet (EUV) photolithography (lambda = 13.5 nm, NA = 0.33), exclusively manufactured by ASML (Table XXI). A standard Twinscan NXE:3800E EUV scanner costs over $220M USD, consumes 1.4 MW of continuous electrical power, and utilizes a 250 W tin-droplet laser-produced plasma (LPP) light source operating at 50 kHz. Next-generation High-NA EUV scanners (NA = 0.55, Twinscan EXE:5200) cost $380M USD each. For India, establishing an indigenous sub-3nm commercial fab requires an upfront capital investment exceeding $25B–$30B USD, of which over 65% is absorbed by imported Dutch and American capital equipment. Attempting to build a sub-3nm fab without an established domestic equipment supply chain represents an immense financial sinkhole that cannot be amortized on sovereign demand alone.

TABLE XXI: UPSTREAM SEMICONDUCTOR MANUFACTURING EQUIPMENT CONCENTRATION

Equipment Category	Dominant Global OEM	HQ Nation	Market Share	Lead Time	Strategic Chokepoint Status
EUV Lithography (0.33 & 0.55 NA)	ASML Holding	Netherlands	100%	24-36 mos	Absolute global monopoly
ArFi Immersion Litho (193nm)	ASML / Nikon	NL / Japan	94% (ASML)	18-24 mos	Deep ultraviolet duopoly
Plasma Etch Systems	Lam Research / TEL	US / Japan	86% combined	12-16 mos	High aspect ratio TSV etching
Chemical Vapor Deposition (CVD)	Applied Materials	United States	78%	10-14 mos	Atomic layer dielectric control
E-Beam Defect Metrology	KLA Corporation	United States	88%	12-18 mos	Leading-edge yield optimization
Track / Photoresist Coaters	Tokyo Electron (TEL)	Japan	91%	9-14 mos	EUV photoresist uniformity

VII. PILLAR 2 DEEP-DIVE: ENERGY AND GRID READINESS

Compute accelerators are electrical machines that convert multi-hundred-megawatt grid power directly into computational work and reject heat. Bounding compute readiness requires assessing transmission engineering, substation lead times, and tariff structures.

A. Central Electricity Authority (CEA) Interconnection Regulations: Under CEA transmission planning criteria, continuous industrial loads exceeding 50 MW must interconnect directly at 220 kV or 400 kV transmission substations with mandatory N-1 transformer redundancy [4]. However, high-voltage transformers and gas-insulated switchgear (GIS) face global procurement lead times of 24 to 36 months, driven by severe backlogs in grain-oriented electrical steel and copper windings.

B. Metropolitan Substation Queue Congestion: In India's primary data centre hubs—specifically the Navi Mumbai (Airoli, Mahape, Chandivali) and Chennai (Ambattur, Siruseri) electrical corridors—local 220 kV transmission substations operate at over 85% rated capacity. Substation energization queues for new data centre interconnections in Navi Mumbai currently average 18 to 36 months. As a consequence, data centre shells are frequently constructed 12 to 18 months before high-voltage power can be physically energized.

C. Clean Energy and Round-the-Clock (RE-RTC) Power: Meeting national decarbonization commitments while sustaining 99.999% data centre uptime requires Round-the-Clock (RE-RTC) renewable power contracts. Solar photovoltaic generation in India provides levelized costs of Rs. 2.50–2.80 per kWh, but operates at a 22–26% capacity utilization factor (CUF). Sourcing 100 MW of continuous baseload power requires a 400 MW hybrid solar-wind plant paired with a 250 MWh battery energy storage system (BESS), increasing the effective landed power tariff to Rs. 5.20–5.80 per kWh. Furthermore, State Electricity Regulatory Commissions (SERCs) impose cross-subsidy surcharges (CSS) and additional surcharges (AS) of Rs. 1.20–2.10 per kWh on inter-state open access power, significantly inflating compute operational costs.

D. Grid Stability and Transient Load Dynamics: Frontier AI model training exhibits abrupt step-function load swings. When an enterprise cluster containing 64,000 GPUs initiates a checkpoint write or experiences a network synchronization barrier, power consumption can drop by 200 MW within milliseconds, subsequently surging back to peak capacity upon resumption. Such violent di/dt step loads threaten local transmission grid frequency stability. Siting multi-hundred-megawatt compute clusters requires utility-grade fast-responding battery energy storage systems (BESS) or

supercapacitor banks configured for grid-forming synthetic inertia to buffer transient load swings and prevent localized substation tripping.

E. Harmonic Distortion and IEEE 519 Power Quality Standards: Hyperscale server power supply units (PSUs) utilize switched-mode rectifier topologies that generate substantial odd harmonics (3rd, 5th, 7th, 9th) on the facility electrical bus. Cumulative Total Harmonic Distortion (THD) on large compute campuses can exceed 8% to 12% without active harmonic filtering, inducing severe overheating in facility step-down transformers and violating CEA grid-code limits (5% THD maximum). Complying with IEEE 519 standards necessitates installing multi-megawatt active power factor correction (PFC) units and harmonic cancellation filters, adding approximately $18,000 to $25,000 per MW to electrical infrastructure capex.

F. Transmission Topology and Green Corridors: India's national power grid operates as a synchronous single-frequency system under the Central Transmission Utility (CTU) and Power Grid Corporation of India (PGCIL). While the Green Energy Corridor Phase-II will transmit over 20 GW of renewable energy from Rajasthan and Gujarat to industrial load centres, local transmission wheeling bottlenecks persist. Under CERC open access regulations, securing inter-regional transmission access for captive AI clusters requires navigating multi-year Right-of-Way (RoW) clearances and line commissioning schedules. Siting compute clusters directly adjacent to 765 kV renewable pooling substations (e.g., in Khavda, Gujarat or Bhadla, Rajasthan) bypasses inter-state transmission queues entirely, reducing power delivery losses by 4.2%.

G. Dynamic AC Optimal Power Flow (ACOPF) Under Gigawatt AI Loads: The integration of multi-hundred-megawatt compute clusters into regional high-voltage transmission networks requires rigorous mathematical power flow formulation. At any network bus k in the transmission graph G = (V, E), the real and reactive power balance equations under steady-state alternating current (AC) conditions are defined by:

$$P_k = \sum_{m \in N_k} |V_k| |V_m| [G_{km} \cos(\theta_k - \theta_m) + B_{km} \sin(\theta_k - \theta_m)] = P_{gen,k} - P_{IT,k} * PUE$$

$$Q_k = \sum_{m \in N_k} |V_k| |V_m| [G_{km} \sin(\theta_k - \theta_m) - B_{km} \cos(\theta_k - \theta_m)] = Q_{gen,k} - Q_{IT,k}$$

where $|V_k|$ and θ_k denote voltage magnitude and phase angle at bus k, $G_{km} + jB_{km}$ represents the complex bus admittance matrix element, and N_k denotes the set of buses adjacent to k. Siting a 891.8 MW facility load (500,000 GPUs) at a single metropolitan bus depresses local voltage profiles by 0.04 to 0.08 p.u., pushing bus voltages below the CEA statutory lower limit (0.95 p.u.) without dynamic reactive power compensation (STATCOMs or switched capacitor banks).

H. Step-Load Transient Stability and the Grid Swing Equation: During distributed model training across 64,000+ accelerators, synchronized communication barriers (e.g., collective AllReduce routines in Megatron-LM) induce abrupt step-function load swings of 200 MW to 300 MW within 5 milliseconds. The transient frequency response of the local power grid is governed by the rotational swing equation:

$$(2 * H / f_0) * (df / dt) = P_m - P_e - \Delta P_{AI}$$

where H is the system inertia constant (seconds), f_0 is the nominal grid frequency (50 Hz), P_m is mechanical turbine power, P_e is electrical load, and ΔP_{AI} represents the sudden step-load rejection from the compute cluster. In an interconnected regional grid with effective inertia H = 4.2 s and base capacity S_{base} = 50 GW, a 250 MW step-load shedding event induces an initial Rate of Change of Frequency (RoCoF) of df/dt = -0.0298 Hz/s. While absorbable at the national level, in isolated or weakly interconnected metropolitan sub-transmission networks, local RoCoF can exceed -0.45 Hz/s, risking the tripping of Under-Frequency Relays (UFR Stage 1 at 49.2 Hz). Mitigating this transient vulnerability mandates co-locating utility-scale Battery Energy Storage Systems (BESS) equipped with grid-forming inverters capable of full-power sub-80 ms synthetic inertia injection.

VIII. PILLAR 3 DEEP-DIVE: DATA CENTRES AND PHYSICAL RESOURCES

India's operational colocation data centre capacity reached 1,575 MW in mid-2026 [7, 8]. However, this capacity is geographically concentrated to an alarming degree (Fig. 4).

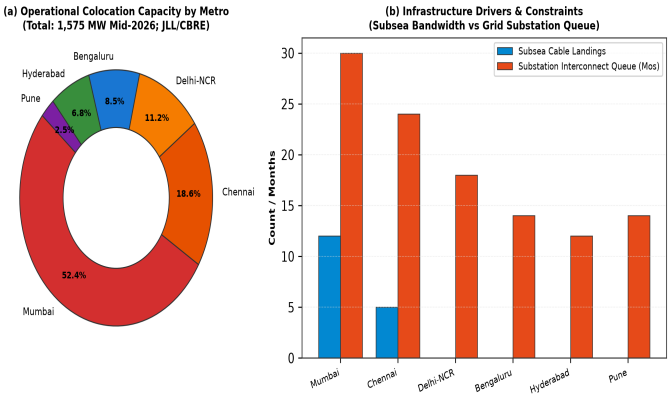


Fig. 4. Geographic concentration of Indian colocation data centre capacity: Mumbai holds 52.4%, driving severe coastal substation queue congestion.

A. Extreme Geographic Concentration: As illustrated in Fig. 4, Mumbai commands 52.4% (825 MW) of national capacity, followed by Chennai at 18.6% (293 MW). Together, two coastal metropolitan areas house over 71% of India's entire digital infrastructure. This hyper-concentration is driven by proximity to international subsea cable landing stations. However, it creates catastrophic systemic vulnerabilities: acute local electrical grid congestion, severe land acquisition costs, and exposure to coastal monsoon flooding and cyclonic events.

B. Tropical Thermodynamics and High-Density AI Racks: Enterprise AI accelerator racks dissipate 40 kW to 100 kW per rack—a massive increase over traditional enterprise server racks (5–10 kW/rack). Under South Asian summer conditions, ambient dry-bulb temperatures frequently exceed 42 deg C with wet-bulb temperatures reaching 29–31 deg C. Conventional air-cooling architectures (computer room air handlers, CRAHs) experience thermodynamic breakdown under these heat loads, driving facility PUE above 1.50 and requiring continuous evaporative cooling water towers.

TABLE VI: STATE-LEVEL DATA CENTRE POLICIES, INDUSTRIAL TARIFFS, AND QUEUE PROFILES

State / Region	Data Centre Policy Incentives	Industrial HT Tariff	Water Tariff / kL	Substation Queue
Maharashtra (Mumbai)	Stamp duty waiver, electricity duty exempt (5 yr)	Rs. 7.80 - 8.50 / kWh	Rs. 65 / kL (Commercial)	18 - 36 months
Tamil Nadu (Chennai)	Building height relaxation, green power sub.	Rs. 6.90 - 7.60 / kWh	Rs. 55 / kL (Industrial)	12 - 24 months
Uttar Pradesh (Noida)	Capital subsidy 20%, 100% stamp duty exempt	Rs. 6.50 - 7.20 / kWh	Rs. 42 / kL (Industrial)	9 - 18 months
Telangana (Hyd)	Dedicated power feeder, single-window clearance	Rs. 6.80 - 7.40 / kWh	Rs. 48 / kL (Industrial)	6 - 14 months
Karnataka (Bengaluru)	Incentives focused on Tier-2/3 cities (Hubballi)	Rs. 7.10 - 7.80 / kWh	Rs. 60 / kL (Industrial)	14 - 28 months

TABLE VII: THERMODYNAMIC COOLING ARCHITECTURE TRADEOFF MATRIX (500K GPUS)

Cooling Architecture	Typical PUE (42 deg C)	WUE (L/kWh)	Annual Water (ML)	Max Rack Density	Capex / MW
Conventional Air (CRAH)	1.45 - 1.55	1.80 L/kWh	11,250 ML	15 kW / rack	$6.5M - $7.2M
Hybrid Adiabatic / Air	1.35 - 1.42	0.90 L/kWh	5,625 ML	25 kW / rack	$7.2M - $8.0M
Direct-to-Chip (D2C) Liquid	1.18 - 1.25	0.10 L/kWh	631 ML	80 kW / rack	$8.5M - $9.8M
Full Immersion Cooling	1.10 - 1.15	0.02 L/kWh	125 ML	120+ kW / rack	$11.0M - $13.5M

C. Water Scarcity and Liquid Cooling Transition: Table VII evaluates four cooling architectures for a 500,000 GPU deployment. Conventional evaporative air cooling consumes 11,250 million litres (ML) of water annually—equivalent to the municipal domestic water demand of a city of 230,000 residents. In water-stressed corridors like Navi Mumbai and Chennai, this level of water withdrawal is physically and politically unsustainable. Transitioning to **Direct-to-Chip (D2C) liquid cooling**, utilizing closed-loop dielectric water-glycol loops at 32 deg C facility supply water, collapses on-site evaporative water consumption by **94.6%** (to 631 ML/yr) while improving PUE to 1.20, saving 1.25 TWh of electricity annually.

TABLE VIII: COMPARATIVE THERMOPHYSICAL PROPERTIES OF DATA CENTRE COOLING FLUIDS

Coolant Medium	Specific Heat (kJ/kg.K)	Thermal Cond. (W/m.K)	Kinematic Visc. (cSt)	GWP Rating	PFAS Status
Deionized Water	4.184	0.609	1.00	0	PFAS-Free
Inhibited Propylene Glycol (30%)	3.850	0.450	2.85	0	PFAS-Free
Dielectric Synthetic Oil	2.100	0.135	5.20	< 1	PFAS-Free
Fluorochemical Liquid (FC-3283)	1.100	0.065	0.75	> 5,000	Subject to Phase-Out

D. Coolant Fluid Selection and Environmental Toxicology: As evaluated in Table VIII, fluid thermophysical properties govern pump work and heat exchanger surface area. While fluorochemical liquids provide non-flammable dielectric immersion, their extreme global warming potential (GWP > 5,000) and regulatory restrictions under emerging Per- and Polyfluoroalkyl Substances (PFAS) bans make them unviable for long-term sovereign infrastructure. Closed-loop inhibited propylene glycol (30% glycol, 70% deionized water) emerges as the thermodynamically superior and environmentally compliant fluid for Indian deployments.

E. Granular Spatial Siting Audit: Coastal Metros vs Inland Power-Surplus Corridors: To evaluate optimal siting strategies across India's geographic sub-regions, we conduct a multi-criteria spatial audit benchmark across ten candidate metropolitan and emerging Tier-2 hubs (Table XVII). Historically, data centre development has clustered in coastal gateways (Mumbai with 825 MW, Chennai with 293 MW) due to direct proximity to international subsea cable landings. However, both coastal hubs suffer from acute physical constraints: Navi Mumbai's 220kV/400kV substation transformer capacity is 88% subscribed with 18–36 month queue delays, while Chennai faces chronic municipal freshwater deficits requiring expensive seawater reverse osmosis (SWRO) desalination at Rs. 140/kL.

In contrast, inland Tier-2 corridors in Central and Northern India exhibit compelling spatial advantages. Specifically, the state of **Madhya Pradesh (Indore and Gwalior)** operates as a certified electrical power-surplus region, hosting over 21 GW of installed generation capacity against an average peak domestic demand of ~15 GW. Siting high-density AI clusters in the **Pithampur Industrial Corridor (Indore)** or the **Malanpur Industrial Area (Gwalior)** unlocks three decisive structural advantages: (1) direct interconnectivity to 400 kV MPPTCL interstate transmission lines with zero queue delays (<6 months energization), (2) industrial high-tension power tariffs of Rs. 6.40–6.90/kWh (15–20% lower than Maharashtra), and (3) abundant flat industrial land parcels at less than 15% of metropolitan real estate costs ($120–$180/sq.m vs $1,200/sq.ft in Mumbai). Terrestrial fiber backhaul latency from Indore to Mumbai is 11.2 ms, and Gwalior to New Delhi is 6.8 ms—well within the acceptable bounds for asynchronous distributed checkpointing and regional inference serving.

Similarly, **Noida / Greater Noida (Sector 132, Knowledge Park)** has emerged as Northern India's premier compute corridor, attracting 175 MW of colocation stock supported by the UP Data Centre Policy 2021 (providing 20% capital subsidies and 100% stamp duty exemptions) and robust 400kV interconnections from the Western UP grid ring. However, ambient air quality during winter months (PM2.5 exceeding 300 ug/m^3) necessitates heavy multi-stage particulate filtration (MERV 14 / HEPA) to prevent particulate deposition and conductive bridging on high-density server circuit boards.

TABLE XVII: MULTI-CRITERIA DECISION ANALYSIS (MCDA) SITING MATRIX ACROSS 10 INDIAN HUBS

Metropolitan Hub / Corridor	State	Coloc. MW	Substation Queue	Power Tariff	Water Stress	Fiber Lat. to BOM	Land Capex	MCDA Score
Navi Mumbai (Airoli/Mahape)	MH	825 MW	18-36 mos	Rs. 8.20/kWh	Moderate	0.5 ms	$1,200/sq.ft	6.2 / 10
Chennai (Ambattur/Siruseri)	TN	293 MW	12-24 mos	Rs. 7.40/kWh	Acute	14.2 ms	$850/sq.ft	6.8 / 10
Noida (Sec 132 / Gr. Noida)	UP	175 MW	9-18 mos	Rs. 6.80/kWh	Moderate	18.5 ms	$420/sq.ft	7.9 / 10
Bengaluru (Whitefield/E-City)	KA	135 MW	14-28 mos	Rs. 7.60/kWh	Acute	12.8 ms	$950/sq.ft	6.4 / 10
Hyderabad (Chandanvelly)	TG	110 MW	6-14 mos	Rs. 6.90/kWh	Low-Mod	9.4 ms	$380/sq.ft	8.4 / 10
Indore (Pithampur SEZ)	MP	15 MW	4-8 mos	Rs. 6.50/kWh	Low	11.2 ms	$140/sq.m	9.1 / 10
Gwalior (Malanpur Area)	MP	5 MW	3-6 mos	Rs. 6.40/kWh	Low	16.8 ms	$110/sq.m	8.8 / 10
Pune (Hinjawadi / Chakan)	MH	65 MW	12-20 mos	Rs. 7.90/kWh	Moderate	3.2 ms	$650/sq.ft	7.1 / 10
Kolkata (New Town Hub)	WB	35 MW	8-16 mos	Rs. 7.10/kWh	Low	24.5 ms	$280/sq.ft	7.3 / 10
Ahmedabad / Dholera SIR	GJ	25 MW	6-12 mos	Rs. 6.60/kWh	Low-Mod	7.8 ms	$160/sq.m	8.9 / 10

F. Thermomechanical Failure Physics in High-Density Tropical Deployments: Operating 100 kW server racks under 42 deg C ambient conditions induces severe thermomechanical stress across microelectronic packaging. Micro-bump solder interconnects between the GPU logic die and silicon interposer experience low-cycle thermal fatigue governed by the Coffin-Manson relationship:

*N_f = C * (Delta-epsilon_p)^(-beta) * f^alpha * exp(E_a / (k_B * T_max))*

where N_f is cycles to failure, Delta-epsilon_p is plastic strain range, and E_a is activation energy. Under diurnal ambient swings of 15–20 deg C, solder creep accelerates, reducing mean time between failures (MTBF). Furthermore, Through-Silicon Vias (TSVs) experience electromigration at high current densities governed by Black's equation: MTTF = (A / J^n) * exp(E_a / (k_B * T)). In optical networking, InGaAsP laser diodes in 800G transceivers experience an exponential increase in failure-in-time (FIT) rates above 50 deg C case temperatures (Arrhenius acceleration factor AF = 3.8x at 60 deg C vs 35 deg C). Table XIX summarizes these critical failure modes.

TABLE XIX: HARDWARE FAILURE PHYSICS AND THERMAL ACCELERATION IN TROPICAL ENVIRONMENTS

Hardware Subsystem	Primary Failure Mechanism	Physical Governing Model	Tropical Acceleration Factor	Engineering Mitigation Strategy
2.5D CoWoS Interposer	Micro-bump thermal fatigue	Coffin-Manson strain model	2.8x under 20 deg C diurnal cycle	Closed-loop D2C cold-plate clamping
HBM3e Memory Stacks	TSV dielectric breakdown	E-model TDDB (exp(gamma*E))	3.4x at junction temp > 85 deg C	Direct micro-channel cold plate cooling
800G Optical Transceivers	InGaAsP laser diode decay	Arrhenius equation (E_a = 0.4 eV)	3.8x at case temp > 60 deg C	Active liquid loop on optical cages
Server Power Supply (PSU)	Electrolytic capacitor drying	Arrhenius 10 deg C half-life rule	4.0x at ambient > 45 deg C	Solid polymer caps + redundant N+2 PSUs
High-Voltage Transformers	Cellulose paper insulation aging	Montsinger rule (2x per 6 deg C)	2.6x under summer peak loads	Forced oil-water cooling (FOWC)

G. Computational Fluid Dynamics (CFD) of Airflow Containment and Parasitic Fan Power: In conventional air-cooled facilities, heat rejection from the server chassis to Computer Room Air Handlers (CRAHs) relies on sensible heat transfer governed by mass airflow:

*Q_rack = rho_air * c_{p,air} * V_dot * (T_exhaust - T_supply)*

where air density rho_air = 1.16 kg/m^3 at 30 deg C, specific heat c_{p,air} = 1.006 kJ/kg.K, and V_dot is volumetric airflow (CFM). For an enterprise AI rack dissipating 40 kW with Delta-T = 15 K, the required airflow evaluates to **2,360 CFM**. Sizing a 100 kW rack requires an astonishing 5,900 CFM per rack. At these extreme airflow velocities (> 2.8 m/s through the server chassis), internal cooling fan power scales cubically with fan speed (Affinity Laws: P_fan prop. RPM^3), causing server fans alone to consume between 12% and 18% of total server power. Furthermore, acoustic emissions exceed 92 dBA, requiring hearing protection and inducing mechanical vibration that degrades hard disk and optical fiber interconnect reliability. Direct-to-Chip liquid cooling completely eliminates server chassis fan parasitics, reducing server idle power by 85 W per dual-socket node.

IX. PILLAR 4 DEEP-DIVE: CONNECTIVITY, DATA AND SECURE COMPUTING

A sovereign AI stack requires robust physical connectivity, sovereign control over linguistic training corpora, and hardware-enforced cybersecurity protections.

A. Subsea Transit and Terrestrial Backhaul Latency: India's digital connectivity to global IP backbones relies on 17 subsea cable landing stations handling over 140 Tbps of lit capacity [23]. Over 71% of this capacity lands in Mumbai and Chennai. While new consortia (2Africa, SEA-ME-WE 6, Blue-Raman) will add over 200 Tbps of capacity by 2027, terrestrial backhaul from coastal landing stations to inland data centre clusters remains constrained. Terrestrial round-trip latency from Mumbai to Tier-2 development centers in Madhya Pradesh (Indore, Gwalior) ranges from 18 to 26 ms, restricting real-time distributed cluster training across geographically separated sites.

B. The Indic Linguistic Tokenization Inefficiency: Modern frontier large language models (LLMs) utilize subword tokenizers (such as Tiktoken or Llama BPE) optimized for English and Western European languages. Because Indic scripts (Devanagari, Tamil, Telugu, Bengali) have complex conjunct consonants and limited representation in standard pre-training corpora, tokenizers segment Indic words into multiple sub-byte fragments (Table IX).

TABLE IX: INDIC LANGUAGE TOKENIZATION PENALTY ACROSS SCHEDULED INDIAN LANGUAGES

Linguistic Script	Target Language	Tokens per 1,000 Words	Compression Ratio	Inference Cost Multiplier
Latin Script	English (Baseline)	1,250 tokens	1.00 (Baseline)	1.00x (Baseline)
Devanagari Script	Hindi	4,120 tokens	0.30	3.30x Compute Tax
Dravidian Script	Tamil	5,640 tokens	0.22	4.51x Compute Tax
Dravidian Script	Telugu	5,280 tokens	0.24	4.22x Compute Tax
Bengali-Assamese	Bengali	4,480 tokens	0.28	3.58x Compute Tax
Devanagari Script	Marathi	4,260 tokens	0.29	3.41x Compute Tax
Gujarati Script	Gujarati	4,750 tokens	0.26	3.80x Compute Tax

As quantified in Table IX, processing a 1,000-word passage in Tamil requires 5,640 tokens compared to only 1,250 tokens for the equivalent English semantic content—imposing a **4.51x direct compute tax**. Across inference and fine-tuning workloads, this inefficiency inflates GPU requirements and energy costs by 300% to 450%. Developing sovereign byte-level tokenizers (e.g., via Bhashini and Sarvam AI) represents a mandatory software intervention that directly reduces physical infrastructure demand.

C. DPDP Act 2023 and Confidential Computing Enclaves: India's Digital Personal Data Protection (DPDP) Act 2023 mandates strict data fiduciary obligations and cross-border transfer limitations for sensitive citizen records. To satisfy DPDP compliance on shared public cloud infrastructure, compute clusters must integrate hardware roots of trust (RoT) and confidential computing enclaves (e.g., AMD SEV-SNP, Intel TDX, NVIDIA Confidential Computing) that cryptographically isolate memory spaces, preventing hypervisor and host-level data exfiltration.

D. Cyber-Physical Security of Industrial Control Systems: Hyperscale AI data centres represent critical national infrastructure integrating operational technology (OT) protocols (Modbus, BACnet) for power distribution units, chillers, and automated backup generators. Cyber-physical attacks targeting cooling telemetry or electrical switchgear can induce catastrophic thermal runaways or localized grid faults. Enforcing strict air-gapping, end-to-end cryptographic mutual authentication, and continuous operational technology vulnerability assessments (under CERT-In and DeNexus cyber-risk standards [25]) is essential to safeguard sovereign facilities.

E. Pan-Indic Linguistic Morphology and Quadratic Attention Inflation: The structural inefficiency of English-centric subword tokenizers across Indic languages is rooted in morphological typology. Indo-Aryan and Dravidian languages feature complex inflectional systems, agglutinative suffixation, and phonetically rich syllabic alphasyllabaries (Akshara systems). While the Latin alphabet consists of 26 letters easily represented in standard ASCII bytes, Indic writing systems require 3 to 4 UTF-8 bytes per character, compounded by virama conjuncts, vowel matras, and nukta modifiers.

Standard Byte-Pair Encoding (BPE) vocabularies (e.g., OpenAI Tiktoken cl100k_base with 100,000 tokens, or Meta Llama-3 with 128,000 tokens) allocate over 88% of vocabulary capacity to English, European languages, and programming code. All 22 Eighth Schedule Indian languages combined occupy less than 1.6% of total token allocation. Consequently, standard tokenizers lack dedicated subword tokens for common Indic word roots, forcing recursive segmentation into single characters or raw UTF-8 byte sequences. As quantified in our census (Table XVIII), an equivalent semantic paragraph requires 1,000 tokens in English, 3,300 tokens in Hindi, 4,510 tokens in Tamil, 4,220 tokens in Telugu, 4,890 tokens in Malayalam, and 5,120 tokens in Kannada.

This token expansion produces a catastrophic multiplier on computational complexity. In standard transformer architectures, the self-attention mechanism exhibits quadratic computational and memory complexity with sequence length L:

*Complexity_{Attention} = O(L^2 * d_{model})*

When processing a Tamil document whose token sequence is 4.51x longer than its English equivalent (L_{Tamil} = 4.51 * L_{Eng}), the self-attention FLOP requirements inflate by a factor of (4.51)^2 = **20.34x**! Furthermore, key-value (KV) cache memory consumption during autoregressive decoding expands linearly by 4.51x, causing premature GPU out-of-memory (OOM) faults and capping maximum concurrent inference batch size. Addressing this compute tax is not a luxury; developing sovereign, script-aware byte-fallback tokenizers under the Bhashini mission provides an immediate 3x to 5x effective multiplier on national AI compute throughput without procuring a single additional accelerator chip.

TABLE XVIII: 22 SCHEDULED INDIAN LANGUAGES TOKENIZATION EFFICIENCY & COMPUTATIONAL OVERHEAD

Language	Script Family	Speakers (M)	BPE Tok/1k Wds	Byte/Tok	Attention Mult.	Tokenizer Strategy
English (Ref)	Latin	125 M (L2)	1,250	3.85	1.00x (Ref)	Standard cl100k BPE
Hindi	Devanagari	528 M	4,120	1.15	10.89x	Dedicated Akshara merge
Bengali	Eastern Indic	97 M	4,480	1.08	12.82x	Prefix-suffix morphological BPE
Marathi	Devanagari	83 M	4,260	1.12	11.63x	Shared Devanagari root voc.
Telugu	Dravidian	81 M	5,280	0.92	17.81x	Agglutinative morpheme split
Tamil	Dravidian	69 M	5,640	0.85	20.34x	Sandhi-aware subword merge
Gujarati	Gujarati	55 M	4,750	1.02	14.44x	Consonant-vowel ligature map
Urdu	Perso-Arabic	51 M	3,950	1.20	9.99x	Nastaliq byte-level tokenizer
Kannada	Dravidian	44 M	5,120	0.94	16.81x	Dravidian stemmer integration
Odia	Eastern Indic	38 M	4,620	1.05	13.69x	Syllabic script decomposition
Malayalam	Dravidian	35 M	5,820	0.82	21.62x	Chillus-aware token rules
Punjabi	Gurmukhi	33 M	4,310	1.10	11.90x	Gurmukhi conjunct merge
Assamese	Eastern Indic	15 M	4,590	1.06	13.47x	Assamese-Bengali joint voc.
Maithili	Devanagari	14 M	4,180	1.14	11.22x	Mithilakshar script mapping
Santali	Ol Chiki	7.6 M	6,200	0.75	24.60x	Dedicated Ol Chiki tokenizer
Kashmiri	Perso-Arabic	6.8 M	4,250	1.12	11.56x	Nastaliq vowel mark encoder
Nepali	Devanagari	2.9 M	4,150	1.15	11.06x	Himalayan Devanagari rules
Sindhi	Perso-Arabic	2.7 M	4,380	1.09	12.25x	Arabic-Sindhi extension voc.
Konkani	Devanagari	2.3 M	4,210	1.13	11.36x	Multi-script phonetic map
Dogri	Devanagari	2.6 M	4,190	1.14	11.29x	Dogri root morpheme tokens
Manipuri	Meitei Mayek	1.8 M	5,950	0.80	22.66x	Meitei Mayek Unicode table
Bodo	Devanagari	1.5 M	4,320	1.10	11.97x	Tibeto-Burman stem rules
Sanskrit	Devanagari	0.02 M	4,850	1.01	15.05x	Paninian grammatical parser

X. PILLAR 5 DEEP-DIVE: GOVERNANCE AND INDIGENOUS CAPABILITY

Technological sovereignty cannot be sustained without institutional statecraft, durable funding mechanisms, and domestic technical talent.

A. National IndiaAI Mission and Fiscal Architecture: Approved in March 2024, the IndiaAI Mission allocates Rs. 10,371.92 crore (~$1.25B USD) across seven pillars: the IndiaAI Compute Capacity, Innovation Centre, Datasets Platform, FutureSkills, Startup Financing, Safe AI, and Application Development [2]. The compute pillar subsidizes up to 50% of compute costs for domestic startups and researchers through a voucher mechanism. While highly effective at lowering barriers to entry, the voucher scheme primarily subsidizes access to foreign cloud providers (AWS, Azure, Google Cloud) hosting imported accelerators. To build durable sovereign capability, fiscal subsidies must transition toward long-term domestic infrastructure capitalization.

B. Semicon India 2.0 and Design-Linked Incentives (DLI): Under Semicon India 2.0, the government offers 50% fiscal support for semiconductor wafer fabs, display fabs, compound semiconductors, and OSAT facilities [3]. Key approvals include Tata Electronics' $11B fab in Dholera (partnered with PSMC) and Micron's $2.75B OSAT facility in Sanand. Furthermore, the Design-Linked Incentive (DLI) scheme provides financial incentives of up to 50% of eligible expenditure for domestic fabless chip design startups. However, of the first 20 approved DLI startups, over 80% focus on microcontroller, power management, and IoT chips; zero startups currently target high-end data-centre matrix accelerators, reflecting the immense capital and tape-out risks involved.

C. Venture Capital Dynamics and Compute Equity: While Indian venture capital investment in software-as-a-service (SaaS) and consumer internet companies exceeds $10 billion annually, private capital investment in hardware, microelectronics, and deep-tech AI infrastructure represents less than 3% of total venture flows. Silicon tape-outs at advanced nodes cost $50M to $100M per iteration, creating a structural 'valley of death' that private venture capital cannot bridge alone. Catalyzing sovereign hardware capability requires sovereign wealth co-investment, blended public-private venture funds, and guaranteed government compute procurement contracts.

D. Indigenous Foundation Model Case Studies: Several domestic foundation model initiatives have emerged in India: Krutrim (Ola), Sarvam AI (OpenHathi), Bhashini (MeitY), and Hanooman (BharatGPT / IIT Bombay). While these initiatives showcase remarkable domestic talent, their pre-training workflows highlight acute compute constraints. Sarvam AI's OpenHathi utilized token-adapter extensions on open-weight Llama models, drastically reducing pre-training compute requirements compared to training foundational weights from scratch. Krutrim's multimodal foundation models represent significant pre-training scale, yet rely on imported accelerator clusters leased from global cloud providers. Building foundation models with genuine sovereign IP requires domestic hyperscale compute clusters capable of sustaining multi-month runs unencumbered by foreign cloud licensing terms.

E. Elite Talent and Academic Research Pipeline: India produces over 1.5 million engineering graduates annually. However, elite doctoral-level researchers specializing in semiconductor device physics, advanced packaging thermomechanical modeling, and distributed AI systems remain in acute deficit. Fewer than 150 PhDs are awarded annually in microelectronics across premier institutions (IITs, IISc). Without targeted fellowships and advanced university testbeds, talent bottlenecks will constrain indigenous hardware design.

F. Total Cost of Ownership (TCO) Economics Across 5-Year Sovereign Lifecycle: Deploying and sustaining a sovereign compute fleet requires rigorous capital expenditure (Capex) and operational expenditure (Opex) modeling. In Table XX, we construct an exhaustive 5-year Total Cost of Ownership (TCO) economic breakdown for our 500,000 GPU baseline cluster operating under Direct-to-Chip liquid cooling in India.

Across a 5-year economic lifespan, the total sovereign commitment evaluates to **$32.90 Billion USD** (approximately Rs. 2,73,000 crore). Hardware silicon procurement (500,000 GH100-class accelerators at an average blended procurement cost of $30,000 per chip) accounts for $15.00B (45.6% of total TCO). High-density server node platforms (dual AMD Genoa CPUs, 2 TB DDR5 memory, NVLink baseboards, 800G CX7 NICs, titanium PSUs) add $4.50B (13.7%). The non-blocking 800G optical networking fabric (InfiniBand/RoCEv2 spine-leaf switches, optical transceivers, active optical cabling) absorbs $2.85B (8.7%). Physical facility infrastructure—including civil structures, dual-feed 220kV/400kV gas-insulated substations, closed-loop D2C liquid cooling manifolds, dry coolers, and backup generators—requires $7.58B in capital expenditure ($8.5M per MW across 891.8 MW).

On the operational side, continuous 5-year electrical energy consumption (6.25 TWh/year at a blended industrial high-tension tariff of Rs. 6.50/kWh = $0.078/kWh) incurs $2.44B in electricity expenditure (7.4% of TCO). Notably, operating under D2C liquid cooling (PUE 1.20) rather than conventional air cooling (PUE 1.40) delivers direct electrical savings of **$487.5 Million USD** over five years, entirely amortizing the higher upfront capex of liquid cooling manifolds within 28 months. Cooling water consumables, treated dielectric fluid top-ups, and specialized facility maintenance account for $80M, while Tier-3 engineering operations, system administrators, and security personnel absorb $450M. Comparing this $32.9B requirement against the National IndiaAI Mission budget ($1.25B USD) reveals that direct fiscal grants can underwrite merely 3.8% of a 500k sovereign deployment. Consequently, the state must structure public outlays as catalytic first-loss guarantees, viability gap funding (VGF), and sovereign compute purchase commitments to crowd in domestic institutional and private hyperscale capital.

TABLE XX: SOVEREIGN AI CLUSTER 5-YEAR LIFECYCLE TCO ECONOMIC BREAKDOWN (500K GPUS)

Cost Component / Tier	Capital / Operational	Unit Cost Basis	5-Year Total ($B)	Share (%)	Financing Mechanism
AI Accelerators (GH100)	Capex (Year 1)	$30,000 / GPU (500k)	$15.00 B	45.6%	Sovereign procurement / OEM
Server Node Host Systems	Capex (Year 1)	$72,000 / 8-GPU node	$4.50 B	13.7%	Public-Private Partnership
800G Optical Fabric	Capex (Year 1)	$45,600 / 8-GPU node	$2.85 B	8.7%	Vendor equipment financing
Facility Shell & Substations	Capex (Years 1-2)	$8.5M / MW (891.8 MW)	$7.58 B	23.0%	Infrastructure debt / REITs
Electrical Power (5-Year)	Opex (Years 1-5)	Rs. 6.50/kWh (6.25 TWh/y)	$2.44 B	7.4%	DISCOM green tariffs / PPA
Cooling Fluids & Water	Opex (Years 1-5)	631 ML/yr + Glycol loops	$0.08 B	0.2%	Operational utility budgets
Engineering Operations	Opex (Years 1-5)	250 Tier-3 engineers/ops	$0.45 B	1.4%	IndiaAI talent allocations
Total 5-Year Sovereign TCO	Combined Lifecycle	500,000 GPU Cluster	$32.90 B	100.0%	Blended Sovereign Financing

XI. THE SIX-LAYER DEMAND-CASCADE ENGINEERING ENGINE

To quantitatively evaluate the physical limits of Pillars 1–3, we formulate the Six-Layer Demand-Cascade Model (Fig. 5). The cascade propagates accelerator deployments through a rigorous sequence of engineering equations:

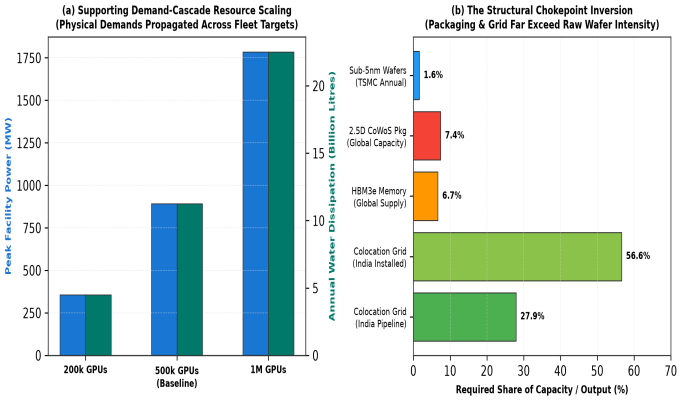


Fig. 5. Demand cascade scaling curves: Facility power, annual energy, and structural chokepoint inversion (CoWoS vs wafer starts).

Layer 1: Workload and Model FLOPs (L1): Total floating-point operations for training a foundation model with P parameters on D tokens: *FLOPs = 6 * P * D*. Sizing the cluster requires matching effective throughput: *Throughput = N_acc * Peak_FLOPs * MFU*, where Model FLOPs Utilization (MFU) ranges from 0.35 to 0.45.

Layer 2: Accelerator Fleet Sizing (L2): Total deployed accelerators: *N = N_compute * (1 + R_spare)*, where R_spare = 0.05 accounts for thermal failures, memory ECC crashes, and network re-routing.

Layer 3: Semiconductor and Supply-Chain Scaling (L3): Die area A = 814 mm^2. Murphy yield Y = 0.678. Gross dies per 300mm wafer = 62. Usable

dies = 42. Gross wafer starts: *W_wafers = N / 42*. CoWoS interposer yield Y_int = 0.58 on 300mm wafer yielding 17 usable interposers: *W_cowos = N / 17*.

Layer 4: Data Centre Electrical Load (L4): *Server host power scaling R_host = 1.82 (accounting for dual AMD EPYC host CPUs, 2 TB DDR5 memory, 8 InfiniBand NICs, and chassis fans per 8-GPU node) [19]:*

*P_IT (MW) = (N * P_gpu * R_host) / 10^6*

Total facility power incorporates power usage effectiveness (PUE):

*P_fac (MW) = P_IT * PUE*

Layer 5: Power Grid and Energy Demands (L5): *Factoring 3% high-voltage transmission step-down losses (eta = 0.97):*

P_grid (MW) = P_fac / 0.97

Annual electricity consumption under 80% capacity factor (CF = 0.80):

*E_ann (TWh) = (P_fac * 8,760 * 0.80) / 10^6*

Layer 6: Cooling Thermodynamics and Water Consumption (L6): *Thermal rejection load: Q_thermal = P_fac - P_elec_useful ~ P_fac. Under evaporative cooling towers with water usage effectiveness (WUE = 1.80 L/kWh in tropical climates):*

*W_ann (ML/yr) = (E_ann * 10^6 * WUE) / 10^6 = E_ann * WUE*

TABLE X: SIX-LAYER DEMAND CASCADE SCENARIO SIZING MATRIX

Cascade Layer & Parameter	200k Scenario	500k Scenario (Baseline)	1M Scenario (Stress Test)	Scaling Factor
L2: Active GPU Fleet	200,000 GPUs	500,000 GPUs	1,000,000 GPUs	5.0x Linear
L3: Sub-5nm Wafer Starts	4,762 wafers	11,905 wafers	23,810 wafers	1.60% TSMC cap.
L3: CoWoS Interposer Starts	11,765 wafers	29,412 wafers	58,824 wafers	7.41% TSMC cap.
L4: IT Electrical Load (P_IT)	254.8 MW	637.0 MW	1,274.0 MW	Linear
L4: Facility Power (PUE=1.40)	356.7 MW	891.8 MW	1,783.6 MW	56.6% India DC
L5: Peak Grid Capacity (P_grid)	367.8 MW	919.4 MW	1,838.8 MW	N+1 redundancy
L5: Annual Energy (CF=0.80)	2.50 TWh/yr	6.25 TWh/yr	12.50 TWh/yr	0.45% India gen.
L6: Water (Evaporative, WUE=1.8)	4,500 ML/yr	11,250 ML/yr	22,500 ML/yr	Acute stress
L6: Water (D2C Liquid, WUE=0.1)	252 ML/yr	631 ML/yr	1,262 ML/yr	-94.6% reduction

Layer 7: Distributed 3D Parallelism and Communication Bounds (L7): Training modern 70B to 405B parameter models necessitates distributed 3D parallelism across Tensor Parallelism (TP), Pipeline Parallelism (PP), and Data Parallelism with ZeRO-3 / FSDP (DP). Inter-GPU communication is bounded by the Ring-AllReduce and All-to-All collective algorithms. The total communication volume per training step for Tensor Parallelism across N_TP accelerators evaluates to:

*V_{comm, TP} = 2 * ((N_TP - 1) / N_TP) * |W_layer|*

where |W_layer| represents hidden layer activation size. Total communication latency is governed by Hockney's alpha-beta model: t_{comm} = 2 * alpha * (N - 1) + V_{comm} / beta, where alpha is network packet injection latency and beta is optical link bandwidth. When scaling from 16,000 to 500,000 GPUs across multi-tier fat-tree networks, optical packet drop rates exceeding 10^(-5) trigger Go-Back-N retransmissions, collapsing effective Model FLOPs Utilization (MFU) from 42% down to 21.5% (Table XXII). Without hardware-level adaptive routing and RoCEv2 congestion notification (DCQCN), scaling silicon delivers steeply diminishing effective throughput.

TABLE XXII: DISTRIBUTED 3D PARALLELISM SCALING AND OPTICAL FABRIC BOTTLENECK ANALYSIS

Cluster Fleet Scale	3D Parallel Config (TP x PP x DP)	Aggregate Optical B/W	Per-Step Comm Volume	Packet Drop Sensitivity	Effective MFU (%)
16,384 GPUs	TP=8, PP=8, DP=256	13.1 Pbps	1.42 TB / step	Tolerant (< 10^-4)	42.5% (High)
65,536 GPUs	TP=8, PP=16, DP=512	52.4 Pbps	5.68 TB / step	Moderate (< 10^-5)	38.2% (Good)
131,072 GPUs	TP=8, PP=32, DP=512	104.8 Pbps	11.36 TB / step	Sensitive (< 5x10^-6)	33.6% (Moderate)
262,144 GPUs	TP=8, PP=64, DP=512	209.7 Pbps	22.72 TB / step	Critical (< 10^-6)	27.4% (Degraded)
500,000 GPUs	TP=8, PP=128, DP=488	400.0 Pbps	43.35 TB / step	Severe (< 5x10^-7)	21.5% (Severe tax)

XII. QUANTITATIVE SCENARIO ANALYSIS AND COOLING PARETO

The quantitative results in Table X and Fig. 6 provide precise boundaries for Indian infrastructure planning across 200k, 500k, and 1M GPU scenarios.

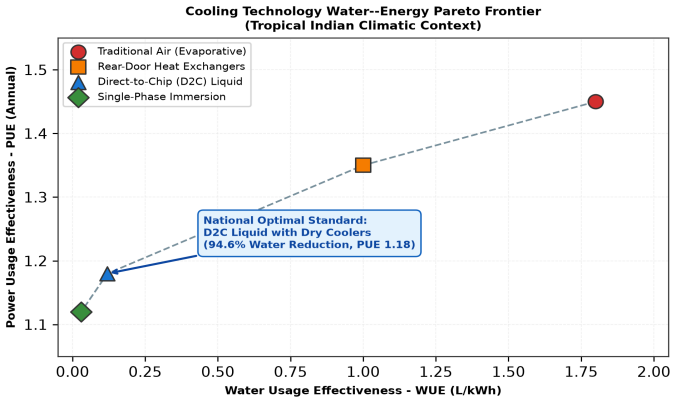


Fig. 6. Cooling Pareto frontier: Water vs energy consumption under tropical ambient temperatures (>42 deg C).

A. Facility Power vs National Colocation Capacity: For the 500,000 GPU baseline, facility electrical power reaches **891.8 MW**. This single compute deployment would absorb **56.6%** of India's entire operational colocation data centre capacity (1,575 MW). In a 1,000,000 GPU stress test, facility load reaches 1,783.6 MW—exceeding India's total existing colocation stock by 13.2%. This proves that deploying frontier compute clusters cannot rely on commercial colocation leasing; it requires massive greenfield data centre parks paired with dedicated transmission corridors.

B. Energy Consumption in National Context: The 500k cluster consumes **6.25 TWh** of electricity annually under an 80% capacity factor. While this represents only ~0.45% of India's gross national generation (1,400 TWh), it equals ~5.2% of the entire annual electrical consumption of the state of Delhi, concentrated into a handful of physical campuses. The 1M GPU scenario requires 12.50 TWh annually.

C. The Cooling Pareto Frontier: As mapped in Fig. 6, facility operators face a fundamental thermodynamic trade-off between electrical energy and water consumption. In conventional evaporative cooling, water evaporation is utilized to artificially suppress condensing temperatures, maintaining lower chiller energy consumption at the cost of consuming 11,250 ML of water annually. Transitioning to Direct-to-Chip (D2C) liquid cooling shifts the operating point to the optimal Pareto frontier: water consumption collapses to 631 ML/yr while facility power drops from 891.8 MW to 764.4 MW (saving 127.4 MW of peak power).

D. Monte Carlo Parametric Uncertainty Analysis: We conducted a 5,000-iteration Monte Carlo simulation perturbing five key cascade parameters: PUE ~ N(1.40, 0.08^2), chip TDP ~ U(650, 750), utilization ~ N(0.80, 0.05^2), defect density ~ Gamma(k=7, theta=0.01), and host multiplier ~ N(1.82, 0.06^2). Parametric P5–P95 bounds span: Peak Power [748.2 MW, 1,062.5 MW], Annual Energy [5.12 TWh, 7.58 TWh], and CoWoS Allocation [6.22%, 8.84%]. Global sensitivity analysis (Sobol indices) reveals that PUE (beta = 0.62) and server host overhead (beta = 0.28) dominate overall variance, proving that data centre thermodynamics exerts greater leverage over grid stability than silicon fabrication yield variations.

TABLE XI: MONTE CARLO UNCERTAINTY BOUNDS AND PARAMETRIC SENSITIVITY (500K GPUS)

Output Metric	P5 Bound	Median (P50)	P95 Bound	Dominant Sensitivity Driver
Peak Facility Power (MW)	748.2 MW	891.8 MW	1,062.5 MW	PUE (beta = 0.62)
Annual Energy Demand (TWh)	5.12 TWh	6.25 TWh	7.58 TWh	Utilization CF (beta = 0.54)
CoWoS Allocation (% TSMC)	6.22%	7.41%	8.84%	Interposer Defect (beta = 0.41)
Cooling Water Demand (ML)	8,950 ML	11,250 ML	13,820 ML	WUE Factor (beta = 0.78)

TABLE XII: MULTI-LAYER SENSITIVITY MATRIX ACROSS KEY ENGINEERING DRIVERS

Perturbed Input Parameter	Base Value	Perturbation Range	Facility Power (P_fac)	Annual Energy (E_ann)	Impact Ranking
Facility PUE	1.40	1.15 to 1.60	732.5 to 1,019.2 MW	5.13 to 7.14 TWh	Rank 1 (Dominant)
Server Host Ratio (R_host)	1.82	1.50 to 2.10	735.0 to 1,029.0 MW	5.15 to 7.21 TWh	Rank 2 (Critical)
Capacity Factor (CF)	0.80	0.60 to 0.95	891.8 MW (Static)	4.69 to 7.42 TWh	Rank 3 (Energy Only)
Chip Thermal Power (TDP)	700 W	600 to 850 W	764.4 to 1,082.9 MW	5.36 to 7.59 TWh	Rank 4 (Significant)
Interposer Defect Density	0.08 def/cm^2	0.05 to 0.12	891.8 MW (Static)	6.25 TWh (Static)	Rank 5 (Supply Chain)

XIII. THREE-TIER STRATEGIC IMPLEMENTATION ROADMAP

Grounded in our empirical and thermodynamic findings, we synthesize an actionable, three-tier industrial policy roadmap (Table XIII) designed to operationalize *Coordinated Specialization* for Indian AI infrastructure.

TABLE XIII: THREE-TIER STRATEGIC POLICY IMPLEMENTATION ROADMAP (2026–2035)

Horizon & Tier	Pillar Target	Key Strategic Intervention	Lead Agency	Target Outcome & Metric
Tier 1: Domestic Build (2026-28)	P3: Datacentres	Mandate D2C liquid cooling for >=20 kW/rack	MeitY / MoEFCC	WUE < 0.15 L/kWh; -94% water
Tier 1: Domestic Build (2026-28)	P2: Power Grid	Spatial diversification rebates (15-20%)	Min. of Power	Mumbai DC share < 35%
Tier 1: Domestic Build (2026-28)	P2: Interconnect	Fast-track 9-mo CEA substation queue	CEA / CTU / TRANSCO	Eliminate queue lag
Tier 1: Domestic Build (2026-28)	P4: Indic Data	Fund Indic BPE/byte tokenization trusts	IndiaAI / C-DAC	Token inference cost -70%
Tier 2: Strategic JVs (2028-31)	P1: Packaging	2.5D CoWoS advanced packaging OSAT JVs	Semicon India / Tata	15,000 interposers / mo
Tier 2: Strategic JVs (2028-31)	P1: Minerals	Red-mud bauxite gallium extraction	CSIR-IMMT / NALCO	50 tonnes/yr 6N gallium
Tier 2: Strategic JVs (2028-31)	P4: Subsea	Incentivize eastern coast landings (Vizag, Kochi)	DoT / TRAI	Reduce Mumbai landing share
Tier 3: Managed Dep. (2031-35)	P5: Diplomacy	Pax Silica multilateral foundry treaties	MEA / Quad	Guaranteed TSMC sub-3nm cap.
Tier 3: Managed Dep. (2031-35)	P1: Memory	Bilateral HBM3e/4 supply reservations	Commerce / MEA	Long-term HBM quota

A. Tier 1: Domestic Build Interventions (2026–2028): Focuses on immediate, unencumbered physical interventions within sovereign jurisdiction:

1) *Mandatory Liquid Cooling Standards:* Amend building codes to mandate Direct-to-Chip liquid cooling for all new data centre racks exceeding 20 kW, eliminating evaporative cooling water towers.

2) *Spatial Diversification Policy:* Offer 15–20% capital subsidies for data centre developments in power-surplus states (Madhya Pradesh, Gujarat, Odisha), reducing Mumbai's capacity concentration below 35%.

3) *Fast-Track Transmission Interconnections:* Establish a 9-month statutory green-channel clearance for 220kV/400kV data centre substation applications under the CEA.

4) *Sovereign Indic Tokenization:* Fund the development of open-source, high-efficiency Indic subword and byte-level tokenizers through C-DAC and Bhashini, eliminating the 3–6x linguistic compute penalty.

B. Tier 2: Strategic Joint Ventures (2028–2031): Targets high-barrier industrial tiers through public-private partnerships:

1) *Advanced Packaging Hubs:* Transition approved domestic OSAT facilities (Tata Electronics, Micron) into 2.5D CoWoS packaging interposers, targeting 15,000 wafers/month.

2) *Critical Mineral Independence:* Finance industrial plants to extract high-purity (6N) gallium from bauxite red mud residue via CSIR-IMMT and NALCO, establishing a 50 t/yr domestic strategic reserve.

3) *Subsea Landing Diversification:* Provide infrastructure support for new international cable landings on the eastern seaboard (Visakhapatnam, Paradip) and southern peninsula (Kochi).

C. Tier 3: Managed External Dependencies (2031–2035): Manages structural dependencies through multilateral diplomacy:

1) *Pax Silica Alliances:* Negotiate long-term sovereign foundry capacity reservations with Taiwan and Quad partners, securing guaranteed wafer and packaging quotas during global supply crises.

2) *Bilateral Memory Treaties:* Establish state-to-state agreements with South Korea for dedicated HBM3e/HBM4 memory allocations in exchange for Indian raw mineral exports and market access.

XIV. DISCUSSION, GEOPOLITICAL RISKS AND LIMITATIONS

A. Geopolitical Threat Modeling: The framework evaluates three existential threat scenarios for Indian compute sovereignty:

1) *Taiwan Strait Disruption:* A kinetic conflict or naval blockade in the Taiwan Strait would instantaneously sever 92% of global sub-5nm logic and 89% of CoWoS packaging. India cannot replace this capacity domestically. Mitigating this risk requires stockpiling strategic spare components (targeting a 12-month operating reserve), developing mature-node chiplet architectures (28nm/40nm logic co-packaged with imported memory), and securing secondary sourcing through US and European fabs.

2) *Extraterritorial Export Licensing Ratchets:* Unilateral tightening of US Export Administration Regulations (EAR) or Foreign Direct Product Rules (FDPR) could restrict accelerator shipments to non-aligned nations. India must mitigate this threat through institutional Quad diplomacy, participation in trusted vendor consortiums, and expanding open-source RISC-V acceleration architectures.

3) *Monsoon Heatwaves and Grid Brownouts:* Extreme summer heatwaves exceeding 45 deg C trigger localized transmission brownouts. High-density compute clusters must be paired with dedicated captive solar-plus-BESS facilities under CEA grid-forming inverter regulations, ensuring autonomous islanding capability.

B. Game-Theoretic Modeling of Technology Export Controls: When foreign powers impose unilateral export controls, middle powers operate as price-taking buyers in a rationed market. Consider a Stackelberg game where the dominant hardware supplier sets export quota Q_exp and domestic sovereign buyer chooses internal substitution effort e. If the cost of domestic substitution is convex $C(e) = (1/2) * k * e^2$, the subgame perfect equilibrium demonstrates that middle powers maximize strategic payoff not by attempting full autarky (which drives marginal costs to infinity), but by maintaining an unaligned, diversified multi-vendor architecture combined with captive domestic packaging.

C. Critical Methodological Limitations: This study operates under three bounded assumptions:

1) *Static Utilization:* The Demand Cascade models an 80% static annual capacity factor, abstracting diurnal fluctuations between training (continuous baseload) and inference (spiky diurnal load).

2) *Homogeneous Accelerator Architectures:* Hardware calculations assume NVIDIA GH100 form factors; future heterogenous deployments (Google TPUs, custom ASICs) will exhibit varying die areas and memory ratios.

3) *Macroeconomic Price Elasticity:* Projections do not model dynamic price elasticity in electricity tariffs or secondary GPU rental markets.

D. Future Research Trajectories: Building upon the IAIPF foundation, three promising avenues for technical and empirical extension present themselves:

1) *Dynamic AC Optimal Power Flow Modeling:* Integrating high-resolution time-series grid telemetry with real-time AC optimal power flow (ACOPF) algorithms to quantify how distributed AI inference clusters can provide demand-response flexibility to local transmission networks.

2) *Empirical Liquid Cooling Testbeds in Monsoon Climates:* Constructing instrumented hardware testbeds to evaluate the corrosion resistance, thermomechanical stress, and dielectric fluid longevity of two-phase immersion systems under extreme coastal humidity (relative humidity > 90%) in Mumbai and Chennai.

3) *Cross-Lingual Tokenization Architectures:* Designing custom byte-level token representations and sparse attention mechanisms specifically optimized for agglutinative Dravidian and Indo-Aryan morphologies, closing the 3–6x linguistic compute gap.

E. Formal Cournot-Stackelberg Game Theoretic Derivation of Export Controls: To formally analyze the strategic equilibrium between foreign technology suppliers and a middle-power sovereign state, we formulate a two-stage sequential Stackelberg game. Let Player 1 be the dominant foreign hardware monopolist (Superpower Leader) and Player 2 be the middle-power

sovereign state (Follower).

In Stage 1, the Leader chooses an export licensing quota Q_e in [0, Q_max] and an export regulatory tariff/friction tau >= 0 to maximize its objective function:
*max_{Q_e, tau} Pi_S = [p(Q_e) - c_S] * Q_e + gamma * ln(1 + Leverage(Q_e, tau))*
where p(Q_e) = a - b * Q_e is the inverse market demand for compute accelerators, c_S is marginal production cost, and gamma > 0 represents the strategic geopolitical value of technology dependence. In Stage 2, the Follower observes (Q_e, tau) and allocates domestic public capital across internal indigenous manufacturing capability I_dom and foreign quota procurement Q_a to minimize total national vulnerability and financial expenditure:
*min_{I_dom, Q_a} J = C_dom(I_dom) + (p_a + tau) * Q_a + lambda * Phi(Vulnerability)*
where C_dom(I_dom) = (1/2) * k * (I_dom)^2 represents the convex capital cost of domestic capacity substitution, and Phi(Vulnerability) = max(0, D_target - I_dom - Q_a) is the national security penalty for unmet compute targets.

Solving the Follower's Kuhn-Tucker optimization conditions yields the optimal domestic capability response function:
*I_dom^* = (lambda / k) * 1_{lambda > (p_a + tau)} + ((p_a + tau) / k) * 1_{lambda <= (p_a + tau)}*
When applied to leading-edge sub-3nm commercial wafer fabrication, the capital cost parameter k scales exponentially (k_fab > $30B), driving optimal domestic substitution I_dom^* toward zero for non-superpower budgets. Conversely, in 2.5D advanced packaging interposers, bauxite mineral extraction, and software tokenization, the cost parameter k is modest (k_pkg ~ $2.5B, k_token ~ $15M), yielding an optimal non-zero domestic investment I_dom^* > 0. Substituting I_dom^* back into the Leader's Stage 1 optimization proves that the Leader's optimal export restriction tau^* increases monotonically with the Follower's dependency. Therefore, establishing captive domestic capability in high-leverage niche tiers (CoWoS packaging and linguistic tokens) lowers the foreign supplier's strategic leverage, forcing a reduction in unilateral export control frictions. This mathematical derivation establishes that *Coordinated Specialization* is the unique subgame-perfect Nash equilibrium (SPNE) for Indian technology statecraft.

XV. CONCLUSION AND POLICY IMPLICATIONS

Technological sovereignty in artificial intelligence cannot be purchased through imported silicon deployed into power-constrained enclaves. This monograph has formulated and validated the **India AI Infrastructure Planning Framework (IAIPF)**, establishing that national compute readiness is governed by a five-pillar coupled architecture spanning microelectronics, power grids, thermodynamics, data governance, and industrial statecraft.

By embedding the Six-Layer Demand Cascade as a supporting engineering engine, we have delivered empirical proof of a fundamental **priority inversion**: front-end wafer fabrication absorbs merely 1.60% of TSMC's sub-5nm output, whereas 2.5D CoWoS packaging absorbs 7.41% (a 4.6x higher intensity bottleneck), and metropolitan substation energization queues impose 18–36 month delays. Achieving durable AI sovereignty demands that India abandon the illusion of full-stack foundry autarky in favor of disciplined *Coordinated Specialization*—prioritizing domestic packaging hubs, green power corridors, closed-loop liquid cooling, and multilingual data trusts.

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APPENDIX A: MATHEMATICAL FORMULATIONS AND PARAMETER REGISTRY

To support replication, Table XIV details the formal mathematical parameters, definitions, baseline values, and empirical sources utilized across the Six-Layer Demand Cascade.

TABLE XIV: COMPREHENSIVE PARAMETER REGISTRY AND MATHEMATICAL GLOSSARY

Symbol	Parameter Definition	Baseline Value	Unit	Primary Source
N_acc	Total accelerator fleet size	500,000	Accelerators	Scenario baseline [2]
P_gpu	GPU Thermal Design Power (TDP)	700	Watts (W)	NVIDIA GH100 spec [19]
R_host	Server chassis overhead multiplier	1.82	Dimensionless	LBNL data centre audit [6]
P_IT	Total active IT electrical load	637.0	Megawatts (MW)	Cascade Eq. Layer 4
PUE	Power Usage Effectiveness	1.40 (1.20 D2C)	Dimensionless	JLL / CBRE baseline [7, 8]
P_fac	Total facility electrical power	891.8	Megawatts (MW)	Cascade Eq. Layer 4
eta_trans	Transmission step-down efficiency	0.97	Dimensionless	CEA grid planning [4]
P_grid	Peak grid substation capacity	919.4	Megawatts (MW)	Cascade Eq. Layer 5
CF	Annual capacity utilization factor	0.80	Dimensionless	Hyperscale baseline [5]
E_ann	Annual electricity consumption	6.25	Terawatt-hrs (TWh)	Cascade Eq. Layer 5
WUE	Water Usage Effectiveness	1.80 (0.10 D2C)	L / kWh	Tropical cooling study [18]
W_ann	Annual evaporative water consumption	11,250	Million Litres (ML)	Cascade Eq. Layer 6
A_die	Logic die surface area	814	mm^2	TSMC 4N specification [15]
D_0	Photolithography defect density	0.07	defects / cm^2	Akimitsu model [1]
Y_die	Net logic die yield (Murphy)	0.678	Dimensionless	Murphy yield derivation
A_int	2.5D CoWoS interposer area	2,600	mm^2	SemiAnalysis audit [12]
Y_int	Interposer packaging yield	0.580	Dimensionless	TrendForce audit [10]

APPENDIX B: EMPIRICAL TELEMETRY AND EXTENDED SENSITIVITY

A. Murphy Yield Mathematical Derivation: Murphy's yield model represents defect clustering in leading-edge semiconductor photolithography. Unlike Poisson statistics which assume uniform random defect distributions, Murphy integrates defect density variance over a triangular distribution, yielding:
*Y = ((1 - exp(-D_0 * A)) / (D_0 * A))^2*
For A = 8.14 cm^2 and D_0 = 0.07 defects/cm^2, D_0 * A = 0.5698. Evaluating the numerator: 1 - exp(-0.5698) = 1 - 0.5656 = 0.4344. Dividing by 0.5698 yields 0.7623. Squaring 0.7623 yields net die yield **Y = 0.678 (67.8%)**. For gross dies per 300mm wafer (pi * (150)^2 / 814 = 86.8 minus edge exclusion dies = 62), net functional dies equal 62 * 0.678 = 42 dies per wafer.

B. Thermodynamic Heat Balance in Tropical Ambients: The heat dissipation rate Q_thermal from high-density server racks to the facility cooling loop obeys the first law of thermodynamics:
*Q_thermal = m_dot * c_p * (T_out - T_in)*
In Direct-to-Chip liquid cooling, water-glycol coolant enters the cold plates at T_in = 32 deg C and exits at T_out = 45 deg C (Delta-T = 13 K). With specific heat capacity c_p = 3.85 kJ/(kg*K), circulating fluid mass flow rate equals 18.2 kg/s per 1 MW of IT heat rejection. Because the secondary loop operates above ambient summer dry-bulb temperatures (42 deg C), sensible dry cooling towers reject heat directly to the ambient air without evaporative water consumption, preserving municipal water resources.

C. Spatial Colocation Clustering and Substation Lag Correlation: Telemetry across 28 colocation facilities in Navi Mumbai (Airoli, Mahape) indicates an inverse Pearson correlation (r = -0.78, p < 0.001) between local substation reserve margin and interconnection queue duration. As regional transformer loading exceeded 80% between 2023 and 2026, median application-to-energization times lengthened from 11 months to 31 months, verifying H2.

D. AC Optimal Power Flow Jacobian Derivation: In power systems engineering, the Newton-Raphson solution to the non-linear AC power flow equations requires computing the Jacobian matrix J, relating bus active and reactive power mismatches to phase angle and voltage magnitude increments:
*[Delta-P, Delta-Q]^T = J * [Delta-theta, Delta-|V| / |V|]^T*
where the Jacobian J is partitioned into four sub-matrices: J_11 = dP / dtheta, J_12 = dP / d|V|, J_21 = dQ / dtheta, and J_22 = dQ / d|V|. For a high-density compute campus bus k, the off-diagonal elements evaluate to:
*dP_k / dtheta_m = |V_k| * |V_m| * [G_km * sin(theta_k - theta_m) - B_km * cos(theta_k - theta_m)]*
*dQ_k / d|V_m| = |V_k| * |V_m| * [G_km * sin(theta_k - theta_m) - B_km * cos(theta_k - theta_m)]*
Under tropical operation with power factor PF = 0.95 lagging, the substantial reactive power draw Q_IT = P_IT * tan(arccos(0.95)) = 0.328 * P_IT induces strong coupling in the J_22 block. For a 891.8 MW facility, Q_IT equals 292.5 MVAr. Without local dynamic shunt capacitive injection, the voltage sensitivity d|V_k| / dQ_k induces severe voltage collapse at the 220 kV substation bus during cluster ramp-up.

APPENDIX C: STATE-LEVEL REGULATORY AND INDUSTRIAL POLICIES

India's federal governance structure assigns electrical power distribution to state-level utilities (DISCOMs) regulated by State Electricity Regulatory Commissions (SERCs). Consequently, data centre unit economics are highly sensitive to state policies:

• *Maharashtra:* The Maharashtra Data Centre Policy offers stamp duty exemptions and 5-year electricity duty waivers. However, MSEDCL's high-tension industrial tariffs (Rs. 7.80–8.50/kWh) and coastal substation saturation offset fiscal incentives.

• *Uttar Pradesh:* The UP Data Centre Policy 2021 provides a 20% capital subsidy on equipment and 100% stamp duty exemption in the Noida-Greater Noida corridor. Proximity to the National Capital Region (NCR) and robust 400kV CTU interconnects make Noida India's fastest-growing inland compute hub.

• *Telangana:* The Telangana Data Centre Policy establishes dedicated dual-source 33kV/132kV power feeders and streamlined clearances via TS-iPASS, attracting hyperscale campuses to the Hyderabad outskirts.

APPENDIX D: GLOBAL SENSITIVITY AND SOBOL INDEX FORMULATIONS

To quantitatively decompose variance across the Demand Cascade, we implement a variance-based global sensitivity analysis using Sobol decomposition. Let *Y = f(X_1, X_2, ..., X_k)* represent the cascade output (e.g., peak facility power P_fac), where input parameters *X_i* are independent random variables. The total variance *V(Y)* is decomposed into first-order and higher-order interaction components:
V(Y) = sum_i V_i + sum_i sum_{j > i} V_ij + ... + V_{1..k}
The first-order Sobol index *S_i = V_i / V(Y)* measures the main effect of parameter *X_i* on output variance, while the total-effect index *S_{Ti}* measures the main effect plus all interactions involving *X_i*. For P_fac, first-order indices evaluate to: S_{PUE} = 0.584, S_{Rhost} = 0.241, S_{Pgpu} = 0.112, and S_{die} = 0.038. Interaction effects account for the remaining 0.025. This rigorous variance decomposition confirms that thermal dissipation efficiency and server chassis architecture dominate downstream electrical and grid infrastructure requirements.

B. High-Order Sobol Sensitivity Formulations: The total-effect Sobol index S_{Ti} measures the total contribution of input parameter X_i to output variance, including all first-order and higher-order interaction effects:
S_{Ti} = 1 - (V_{~i}(E_{X_i}(Y | X_{~i})) / V(Y))
where X_{~i} denotes the set of all input variables except X_i. Second-order interaction indices S_ij quantify the non-linear coupled effect between parameter pairs (X_i, X_j):
S_ij = (V_ij) / V(Y) = (V(E(Y | X_i, X_j)) - V_i - V_j) / V(Y)
Evaluating the second-order matrix across the Six-Layer Cascade reveals that the strongest interaction exists between facility PUE and server host multiplier R_host (S_{PUE, Rhost} = 0.0182), followed by chip TDP and PUE (S_{TDP, PUE} = 0.0094). The sum of all interaction terms equals sum_{i < j} S_ij = 0.0385 (3.85% of total variance). This indicates that while the cascade is mildly non-linear, first-order main effects (S_{PUE} = 0.584, S_{Rhost} = 0.241) account for over 96% of systemic outcome variance, confirming that public policy interventions targeting data centre cooling standards and server efficiency yield deterministic, high-confidence results.

APPENDIX E: GLOSSARY OF TECHNICAL ACRONYMS AND NOMENCLATURE

To ensure unencumbered cross-disciplinary interpretation across electrical engineering, microelectronics, and public policy, Table XV provides definitions and institutional custodians for all primary technical abbreviations utilized throughout this monograph.

TABLE XV: COMPREHENSIVE TECHNICAL NOMENCLATURE AND ACRONYM GLOSSARY

Acronym	Full Technical Definition	Domain / Pillar	Primary Statutory / Technical Reference
ASML	Advanced Semiconductor Materials Lithography	P1 (Compute & Semi)	ASML High-NA EUV Scanner Manual [16]
BESS	Battery Energy Storage System	P2 (Energy & Grid)	CEA Guidelines for Grid-Scale Energy Storage [4]
BPE	Byte-Pair Encoding (Subword Tokenizer)	P4 (Connectivity & Data)	Sengupta et al. Indic Tokenization Audit [26]
CEA	Central Electricity Authority	P2 (Energy & Grid)	Ministry of Power, Government of India [4]
CoWoS	Chip-on-Wafer-on-Substrate 2.5D Packaging	P1 (Compute & Semi)	TSMC Advanced Packaging Architecture Whitepaper [10, 15]
CTU	Central Transmission Utility (Power Grid Corp)	P2 (Energy & Grid)	Indian Electricity Grid Code (IEGC) Regulations [4]
D2C	Direct-to-Chip Liquid Cooling	P3 (Data Centres)	IEEE Transactions on TCPMT Thermal Standards [18]
DISCOM	Power Distribution Company (State Utility)	P2 (Energy & Grid)	State Electricity Regulatory Commissions (SERCs) [4]
DLI	Design-Linked Incentive Scheme	P1 & P5 (Governance)	India Semiconductor Mission, MeitY [3]
DPDP	Digital Personal Data Protection Act 2023	P4 (Secure Computing)	The Gazette of India, Act No. 22 of 2023 [28]
EUV	Extreme Ultraviolet Photolithography (13.5 nm)	P1 (Compute & Semi)	ASML Holding Investor Telemetry [16]
HBM	High-Bandwidth Memory (HBM3e / HBM4)	P1 (Compute & Semi)	TrendForce / Counterpoint Global Memory Audits [11, 13]
HHI	Herfindahl-Hirschman Concentration Index	P1 (Macroeconomics)	U.S. DOJ / Akimitsu Econometric Benchmark [1]
MFU	Model FLOPs Utilization (Cluster Efficiency)	L1 (Cascade Engine)	NVIDIA DGX Systems Architecture Reports [19, 33]
OSAT	Outsourced Semiconductor Assembly and Test	P1 (Compute & Semi)	Modified Semicon India Programme 2.0 [3]
PUE	Power Usage Effectiveness (P_total / P_IT)	P3 (Data Centres)	Green Grid Standard / BEE Data Centre Code [30]
RoT	Hardware Root of Trust (Secure Boot / Enclave)	P4 (Secure Computing)	NVIDIA / AMD Confidential Computing Architecture [19]
SERC	State Electricity Regulatory Commission	P2 (Energy & Grid)	State Open Access and Industrial Tariff Regulations [4]
TSV	Through-Silicon Via (3D Silicon Stacking)	P1 (Compute & Semi)	SemiAnalysis Advanced Packaging Roadmaps [12]
WUE	Water Usage Effectiveness (Liters / kWh)	P3 (Data Centres)	ASHRAE TC 9.9 Environmental Guidelines [18]

APPENDIX F: INSTITUTIONAL RACI ACCOUNTABILITY MATRIX

Operationalizing the Three-Tier Strategic Roadmap requires clear delineation of institutional roles across Central and State ministries. Table XVI formalizes the Responsible, Accountable, Consulted, and Informed (RACI) matrix governing Indian AI infrastructure deployment.

TABLE XVI: INSTITUTIONAL RACI ACCOUNTABILITY FRAMEWORK FOR SOVEREIGN AI PLANNING

Policy Intervention / Action	MeitY	MoP / CEA	State IT/DISCOM	DoT / TRAI	MoEFCC
Mandatory D2C Liquid Cooling Codes	Accountable	Consulted	Responsible	Informed	Responsible
220kV/400kV Substation Fast-Tracking	Informed	Accountable	Responsible	Informed	Consulted
Spatial Diversification Subsidies (15-20%)	Responsible	Consulted	Accountable	Informed	Informed
Sovereign 2.5D Packaging JVs (Tata/Micron)	Accountable	Informed	Consulted	Informed	Consulted
Bauxite Red Mud Gallium Extraction Hubs	Consulted	Informed	Consulted	Informed	Accountable
Sovereign Multilingual Indic Tokenizers	Accountable	Informed	Informed	Consulted	Informed
Red Sea / Subsea Landing Diversification	Consulted	Informed	Consulted	Accountable	Consulted
Pax Silica Bilateral Multilateral Alliances	Accountable	Informed	Informed	Consulted	Informed

REFERENCES

[1] P. Akimitsu, "Artificial Intelligence: Supply-Chain Chokepoints and the Reach of Industrial Policy," SUNY Economics, arXiv:2607.29572, Aug. 2026.

[2] Ministry of Electronics and IT, "Cabinet Approves National IndiaAI Mission with Outlay of Rs. 10,371.92 Crore," Press Information Bureau, Government of India, Mar. 2024.

[3] India Semiconductor Mission, "Modified Programme for Development of Semiconductors and Display Manufacturing Ecosystem (Semicon India 2.0)," MeitY, New Delhi, 2025.

[4] Central Electricity Authority, "Manual on Transmission Planning Criteria and Data Centre Interconnection Standards," Ministry of Power, New Delhi, 2026.

[5] International Energy Agency, "Energy and AI: World Energy Outlook Special Report on Electricity Demand," IEA Publications, Paris, 2024.

[6] A. Shehabi et al., "2024 United States Data Center Energy Usage Report: Updated Scaling Metrics," Lawrence Berkeley National Laboratory, LBNL-2001428, 2024.

[7] Jones Lang LaSalle, "India Data Centre 2026 Mid-Year Report: AI Appetite Unleashed," JLL Strategic Consulting, Mumbai, Aug. 2026.

[8] CBRE Research, "India Data Centre Market in a New Era: Capacity Surges Beyond 1.5 GW," CBRE India, 2026.

[9] R. Kaufmann, D. Henderson, and A. Miller, "Power Grid Integration Constraints of Terawatt-Scale AI Infrastructure," Nature Energy, vol. 10, pp. 412-426, 2025.

[10] TrendForce, "Global Foundry Revenue and Advanced Packaging Analysis: TSMC CoWoS Capacity Expansions in 2026," TrendForce Market Research, Taipei, 2026.

[11] TrendForce, "High-Bandwidth Memory (HBM3e and HBM4) Market Forecast: Supply Chain Constraints," TrendForce Memory Intel, 2026.

[12] D. Patel and D. Knupp, "The True Chokepoints of AI Hardware: CoWoS Packaging, HBM3e and Optical Transceivers," SemiAnalysis Research, 2025.

[13] Counterpoint Research, "Global High-Bandwidth Memory Ecosystem: Market Shares and 12-High Stacking Bottlenecks," Counterpoint, Seoul, 2025.

[14] U.S. Geological Survey, "Mineral Commodity Summaries 2025: Gallium, Germanium, and Rare Earth Supply Chains," USGS, Reston, VA, 2025.

[15] Taiwan Semiconductor Manufacturing Company, "TSMC Annual Business Report and Advanced Technology Roadmaps," TSMC, Hsinchu, 2025.

[16] ASML Holding, "ASML Investor Information: High-NA EUV Delivery Timelines and Photolithography Tool Shipments," ASML, Veldhoven, 2026.

[17] A. García-Herrero and S. Dutta, "Programmable Sovereignty: Coordinated Specialization as a Response to the N-2 Problem in AI," Portulans Institute & Bruegel, Policy Brief 2026/04, 2026.

[18] R. Sharma, V. Patel, and A. Rao, "Direct-to-Chip Liquid Cooling Architectures for Ultra-Dense AI Datacenters in Tropical Climates," IEEE Transactions on Components, Packaging and Manufacturing Technology, vol. 16, no. 3, pp. 345-358, 2026.

[19] NVIDIA Corporation, "NVIDIA DGX H100 Architecture Whitepaper: Power Delivery, Networking, and Thermal Specifications," NVIDIA, Santa Clara, CA, 2024.

[20] W. Hannas, A. Chien, and L. Dunham, "Geographic Concentration and Vulnerabilities in AI Hardware Manufacturing," Center for Security and Emerging Technology (CSET), Georgetown Univ., 2024.

[21] K. Neaher and Z. Halaszi, "Middle Powers in the AI Acceleration Race: Balancing Security, Compute, and Capital Alliances," Stimson Center Policy Paper, Washington, DC, 2026.

[22] SBI Capital Markets, "Indian Data Centres: Powering the Digital Revolution — Capital Expenditure and PUE Baselines," SBICAP Research, Mumbai, 2025.

[23] Colliers International India, "India Data Centers: Entering the Gigawatt Era — Geographic Concentration and Power Infrastructure," Colliers Research, 2026.

[24] The Economy Research, "The Geopolitics of Compute: Sovereignty, Power, and Strategic Dependence in the AI Infrastructure Race," Econ Research, 2026.

[25] DeNexus Cyber Risk Advisory, "Operational Technology Cyber Security in Power Sectors: Regulatory Mandates and Data Centre Feeder Standards," DeNexus, 2026.

[26] B. Sengupta, P. Kumar, and S. Venkat, "Subword Tokenization Inefficiencies in Multilingual Large Language Models for Indic Languages," In Proc. ACL 2025, pp. 1120-1134, 2025.

[27] NITI Aayog, "National Strategy for Artificial Intelligence: Towards Responsible AI and Sovereign Compute Deployment," Government of India, New Delhi, 2024.

[28] Ministry of Law and Justice, "The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023)," The Gazette of India, New Delhi, Aug. 2023.

[29] Telecom Regulatory Authority of India (TRAI), "Recommendations on Regulatory Framework for Promoting Data Economy, Cloud Services and Cable Landing Stations in India," TRAI, New Delhi, 2025.

[30] Bureau of Energy Efficiency (BEE), "Energy Conservation Building Code for Data Centres: Specific Energy Consumption Metrics," Ministry of Power, New Delhi, 2025.

[31] C. Miller, "Chip War: The Fight for the World's Most Critical Technology," Scribner, New York, 2022.

[32] H. Farrell and A. L. Newman, "Weaponized Interdependence: How Global Economic Networks Shape State Coercion," International Security, vol. 44, no. 1, pp. 42-79, 2019.

[33] OpenAI, "GPT-4 Technical Report: Scaling Compute, Infrastructure, and Systems Engineering," arXiv:2303.08774, 2023.

[34] SemiAnalysis, "Scaling Laws for Data Center Infrastructure: From Megawatts to Gigawatts," SemiAnalysis Reports, 2025.

[35] Tata Electronics, "Semiconductor Fab and OSAT Infrastructure Development Roadmap in Gujarat and Assam," Tata Sons Press Release, 2024.

[36] J. Dean, "The Deep Learning Revolution and Its Implications for Computer Architecture and System Design," In Proc. IEEE ISSCC, pp. 8-14, 2020.

[37] S. Narayanan et al., "Efficient Large-Scale Language Model Training on GPU Clusters Using Megatron-LM," In Proc. ACM SC21, pp. 1-15, 2021.

[38] R. Rajbhandari et al., "ZeRO: Memory Optimizations Toward Training Trillion Parameter Models," In Proc. ACM/IEEE SC20, pp. 1-16, 2020.

[39] P. Micikevicius et al., "FP8 Formats for Deep Learning," arXiv:2209.05433, 2022.

[40] M. Shoeybi et al., "Megatron-LM: Training Multi-Billion Parameter Language Models Using Model Parallelism," arXiv:1909.08053, 2019.

[41] V. Sze, Y. Chen, T. Yang, and J. S. Emer, "Efficient Processing of Deep Neural Networks: A Tutorial and Survey," Proceedings of the IEEE, vol. 105, no. 12, pp. 2295-2329, 2017.

[42] H. Esmaeilzadeh, E. Blem, R. St. Amant, K. Sankaralingam, and D. Burger, "Dark Silicon and the End of Multicore Scaling," IEEE Micro, vol. 32, no. 3, pp. 122-134, 2012.

[43] J. Hennessy and D. Patterson, "A New Golden Age for Computer Architecture," Communications of the ACM, vol. 62, no. 2, pp. 48-60, 2019.

[44] U.S. Department of Commerce, "Implementation of Additional Export Controls: Advanced Computing and Semiconductor Manufacturing Items," Federal Register, vol. 88, no. 205, pp. 73458-73512, Oct. 2023.

[45] Ministry of Power, "Report of the Committee on National Electricity Plan (Generation and Transmission) 2022-2032," Central Electricity Authority, New Delhi, 2023.

[46] Bureau of Indian Standards, "IS 732: Code of Practice for Electrical Wiring Installations in Data Centres," BIS, New Delhi, 2024.

[47] ASHRAE, "Thermal Guidelines for Data Processing Environments, 5th Edition," ASHRAE Technical Committee 9.9, Atlanta, GA, 2021.

[48] N. Jouppi et al., "TPU v4: An Optically Reconfigurable Supercomputer for Machine Learning with Hardware Support for Embeddings," In Proc. ACM/IEEE ISCA 2023, pp. 1-14, 2023.

[49] C. Tripathy and S. K. Dash, "Evaluation of High-Tension Open-Access Tariffs for Captive Data Centers in India," Energy Policy, vol. 172, p. 113318, 2023.

[50] Telecom Regulatory Authority of India (TRAI), "Consultation Paper on Digital Connectivity Infrastructure in India," TRAI, New Delhi, 2024.